PHINMA UNIVERSITY OF ILOILO

College of Information Technology Education

Calma Transportation Services: Implementation of a Cash Register and Reservation System

---

In partial fulfillment of the requirements in Capstone 1 and Research Subject

Maulion, John Welter

Moises, John Michael

Providio, Michael Harold

Cajayon, Renzo

Ramo, Carel Dame

TABLE OF CONTENTS

CHAPTER 1 – INTRODUCTION …….. 5

1.1 Background of the Study ……..5

1.2 Statement of the Problem ……15

1.3 Objectives of the Study ……17

1.3.1 General Objective ……..17

1.3.2 Specific Objectives ……..18

1.4 Scope and Delimitation ……19

1.5 Significance of the Study ……21

1.6 Definition of Terms ……..22

CHAPTER 2 – REVIEW OF RELATED LITERATURE AND STUDIES …….26

2.1 Foreign Studies ……..26

2.2 Local Studies ……51

2.3 Related Literature ……66

2.4 Synthesis of the Reviewed Literature ……77

CHAPTER 3 – METHODOLOGY ……80

3.1 Research Design ……..80

3.2 System Development Methodology ……..…..82

3.3 System Architecture ……87

3.4 Tools and Technologies Used ……..91

3.5 Data Gathering Techniques ……96

3.6 Testing Procedures ……..99

3.7 Evaluation Criteria ……103

CHAPTER 4: SYSTEM DEVELOPMENT, IMPLEMENTATION, TESTING, EVALUATION, RESULTS, AND DISCUSSION ……..115

4.1 System Workflow Diagram ……...115

4.2 Overview of the Developed System ……...115

4.3 System Development Methodology ……...117

4.4 System Architecture ……121

4.5 Hardware and Software Requirements ……124

4.5.1 Hardware Requirements ……125

4.5.2 Software Requirements ……...125

4.6 Database Design ……..126

CHAPTER 5: CONCLUSION AND RECOMMENDATIONS ……..129

5.1 Introduction ……129

5.2 Summary of Findings ……..129

5.3 Conclusion ……130

5.4 Recommendations ……...131

5.5 Impact of the Study ……134

5.6 Summary ……...135

PHINMA UNIVERSITY OF ILOILO

College of Information Technology Education

CHAPTER 1

INTRODUCTION

1.1 Background of the Study

Transportation has always played a fundamental role in human society, serving as one of the essential systems that enable the movement of people, goods, and services from one location to another in an efficient, organized, and reliable manner. It is a critical component of economic development and social mobility, as it connects individuals to workplaces, educational institutions, healthcare services, and other essential destinations. In today's rapidly evolving and highly connected world, transportation has become even more significant due to the increasing demand for mobility, accessibility, and convenience. Modern lifestyles require faster and more flexible transportation options, especially in urbanized and tourism-driven areas where mobility is no longer considered a luxury but a daily necessity.

In particular, the demand for convenient and reliable transportation services continues to grow among various groups of individuals, including tourists, business professionals, students, and local residents. The increasing movement of people across cities and regions has created a need for transportation solutions that are both flexible and cost-effective. Many individuals prefer temporary access to vehicles rather than owning one due to financial constraints, maintenance costs, insurance expenses, and other long-term responsibilities associated with vehicle ownership. As a result, car rental services have emerged as a practical and efficient solution that allows users to access transportation based on their specific needs, schedules, and duration of use.

Car rental services provide a wide range of benefits such as flexibility, affordability, and convenience, making them an essential part of modern transportation systems. These services are particularly valuable in areas with high tourism activity, where visitors require short-term transportation to explore destinations, attend business meetings, participate in events, or travel between locations efficiently. Likewise, local residents may also rely on rental services for special occasions such as family trips, emergencies, or temporary vehicle replacement. Because of these growing demands, car rental businesses must ensure that their operations are well-organized, efficient, and capable of handling increasing customer requirements.

Car rental businesses operate by allowing customers to reserve and use vehicles for a specific period of time based on their transportation needs. This process typically involves vehicle selection, reservation scheduling, payment processing, and record management. While these steps may appear simple, they require accurate coordination to ensure that vehicles are properly allocated and available when needed. However, as customer demand increases, managing these operations becomes more complex and challenging, especially when handled manually or through traditional methods.

Efficient management of reservations, vehicle availability, customer records, and payment transactions is critical to the success and sustainability of any car rental business. Without proper systems in place, these processes can easily become disorganized, time-consuming, and prone to human error. Businesses must ensure that all transactions are properly recorded, updated, and monitored in real time to maintain smooth operations and high levels of customer satisfaction. Failure to implement an efficient system may result in operational inefficiencies, delayed services, booking conflicts, customer dissatisfaction, and potential financial losses.

Calma Transportation Services is a car rental business that provides vehicle booking and reservation services to both local customers and tourists. The company plays an important role in offering accessible and practical transportation solutions within its service area. It caters to individuals who require reliable transportation services for short-term use, ensuring that customers can travel conveniently without the need for vehicle ownership. However, despite the value of its services, the current operational process of Calma Transportation Services is primarily based on manual methods of recording and managing transactions.

At present, information such as customer bookings, vehicle availability, and payment records are typically documented using handwritten notes, logbooks, and basic record keeping systems. While these traditional methods may have been sufficient during the early stages of the business, they are no longer efficient in handling increasing transaction volumes. Manual processes require more time and effort, and they lack the ability to provide real-time updates, which are essential for modern business operations.

One of the major challenges of manual processing is the difficulty in maintaining accurate, consistent, and well-organized records. Handwritten entries are prone to human errors such as incorrect encoding, misinterpretation of information, and duplication of records. These issues become more frequent when multiple transactions occur at the same time, increasing the risk of booking conflicts where a single vehicle may be assigned to multiple customers. Such problems can cause delays, inconvenience, and dissatisfaction among clients, ultimately affecting the reputation of the business.

In addition, physical records are highly vulnerable to damage, loss, and misplacement. Unlike digital systems that store data securely in databases, manual records depend entirely on physical storage such as paper documents and logbooks. These materials can be easily affected by environmental factors such as moisture, fire, or physical wear and tear. Once records are lost or damaged, retrieving accurate information becomes difficult or even impossible, which disrupts business continuity and operational efficiency.

Another critical issue associated with manual systems is the complexity and inaccuracy of payment processing. Staff members are required to manually compute rental fees, additional charges, and discounts, which increases the likelihood of miscalculations. Even small errors in billing can lead to financial discrepancies, customer disputes, and loss of trust in the business. Moreover, generating reports for daily, weekly, or monthly transactions becomes a time-consuming process since data is not stored in a centralized and easily accessible system.

As technology continues to advance, many businesses across different industries are shifting from traditional manual processes to computerized and web-based systems. Digital transformation has become a key factor in improving operational efficiency, reducing human error, and enhancing data management capabilities. In

the context of transportation and car rental services, computerized systems offer significant advantages such as real-time updates on vehicle availability, automated reservation processing, faster transaction handling, and improved financial accuracy through integrated cash register systems.

A web-based system, in particular, provides additional benefits in terms of accessibility, flexibility, and scalability. Through a web browser, authorized users such as staff and administrators can access the system anytime and anywhere, allowing for more convenient management of daily operations. This eliminates the limitations of traditional systems that require physical presence or manual checking of records. Furthermore, data is securely stored in a centralized database, ensuring that all records are properly organized, consistently updated, and easily retrievable when needed.

The integration of a reservation system with a cash register module further enhances the functionality of the business. This combination allows seamless coordination between booking management and payment processing, ensuring that all transactions are accurately recorded and properly monitored. It reduces redundancy, minimizes human error, and improves transparency in financial operations. In addition, it enables the generation of detailed reports that can be used for monitoring business performance, identifying trends, and supporting strategic decision-making.

The rapid advancement of information and communication technology has significantly transformed the way businesses operate across various industries. Organizations are increasingly adopting computerized systems to improve efficiency, accuracy, and accessibility in managing their daily operations. The transportation and vehicle rental industry is among the sectors that have greatly benefited from technological innovations. Through the implementation of web-based systems, businesses can automate routine processes, reduce paperwork, and provide faster services to customers. As technology continues to evolve, companies that embrace digital transformation gain a competitive advantage by improving service quality and operational performance.

In today's highly competitive business environment, customer satisfaction has become one of the most important factors influencing organizational success. Customers expect services that are fast, reliable, and accessible at any time. Traditional methods of managing reservations and transactions often fail to meet these expectations because they involve manual procedures that consume time and are prone to errors. As a result, many businesses are investing in digital solutions that allow customers to conveniently access services through online platforms. These systems improve communication between service providers and customers while enhancing the overall user experience.

The growing dependence on digital technologies has encouraged organizations to modernize their data management practices. Information serves as a valuable resource that supports decision-making and strategic planning. Businesses that maintain accurate and organized records are better positioned to monitor performance, identify trends, and respond to customer needs effectively. In contrast, organizations that continue to rely on manual record-keeping often experience challenges related to data inconsistency, information loss, and delayed reporting. Therefore, implementing a centralized information system is essential for ensuring the accuracy, security, and accessibility of business data.

The use of online reservation systems has become increasingly common in service-oriented industries because of their ability to streamline booking procedures and improve operational efficiency. Reservation systems provide customers with the convenience of checking availability, selecting services, and confirming bookings without the need for physical interaction. These systems also allow businesses to manage reservations more effectively by reducing scheduling conflicts and ensuring that resources are allocated appropriately. As customer expectations continue to rise, the adoption of reservation technologies becomes a necessity rather than an option for many organizations.

Financial management is another critical aspect of business operations that can benefit from automation. Accurate recording of transactions is essential for monitoring income, controlling expenses, and maintaining financial accountability. Cash register systems play a vital role in ensuring that all transactions are properly documented and easily retrievable when needed. By integrating financial management features into business

systems, organizations can improve transparency, reduce errors, and generate accurate reports that support informed decision-making. The integration of cash register functionality with reservation management can further enhance operational efficiency by consolidating multiple processes into a single platform.

The emergence of web-based technologies has significantly changed the way businesses interact with their customers. Unlike traditional systems that require physical presence, web-based platforms allow users to access services remotely using computers or mobile devices connected to the internet. This level of accessibility provides greater convenience for customers while enabling businesses to extend their reach beyond geographical limitations. Furthermore, web-based systems facilitate real-time communication and information updates, ensuring that users receive accurate and timely information regarding available services and transactions.

As businesses continue to adapt to the digital era, the importance of system integration becomes increasingly evident. Integrated systems enable different business functions to work together seamlessly, reducing redundancy and improving efficiency. In the context of vehicle rental operations, integrating reservation management, customer records, vehicle monitoring, and financial transactions into a unified system can significantly enhance productivity. Such integration minimizes the need for repetitive data entry, improves information consistency, and supports more effective operational control.

The increasing demand for efficient transportation services has also contributed to the growth of vehicle rental businesses. Customers often seek flexible transportation options that provide convenience without the long-term responsibilities associated with vehicle ownership. As a result, car rental companies must continuously improve their services to meet customer expectations and remain competitive in the market. Implementing a modern information system allows rental businesses to respond more effectively to customer demands while ensuring efficient management of available resources.

The development of a Car Rental with Cash Register System aligns with the ongoing trend of digital transformation in the transportation industry. By utilizing modern technologies, the proposed system aims to improve reservation management, transaction processing, and record maintenance. The integration of these functionalities into a single platform can enhance operational efficiency, improve customer satisfaction, and

support accurate financial management. Through automation and centralized data management, the system can address many of the challenges commonly associated with traditional rental operations.

The proposed system is expected to provide long-term benefits for both customers and administrators. Customers will experience a more convenient and efficient method of accessing rental services, while administrators will benefit from improved data organization and streamlined operations. Furthermore, the system can serve as a valuable tool for supporting business growth by providing accurate information and reliable reporting capabilities. As technology continues to shape modern business practices, the adoption of integrated information systems becomes increasingly important in achieving sustainable operational success.

The increasing dependence of businesses on digital technologies has transformed the manner in which services are delivered to customers. In recent years, organizations have recognized the importance of adopting computerized systems to remain competitive in rapidly changing markets. Technology has become an essential tool for improving operational efficiency, reducing human errors, and enhancing customer satisfaction. Businesses that utilize digital solutions are often able to provide faster, more accurate, and more reliable services compared to those that continue to rely on traditional manual procedures. As a result, information systems have become an integral component of modern business operations across various industries, including transportation and vehicle rental services.

The transportation sector plays a vital role in supporting economic growth and social mobility. People rely on transportation services for daily commuting, business activities, tourism, and other essential purposes. With the continuous increase in travel demands, vehicle rental businesses have emerged as a practical alternative for individuals who require temporary transportation without the financial burden of purchasing a vehicle. Car rental services provide flexibility and convenience, making them a popular option among tourists, business travelers, and local residents. Consequently, the demand for efficient and reliable rental management systems has increased significantly as businesses seek to accommodate growing customer expectations.

Many car rental businesses continue to face challenges related to reservation management, transaction recording, vehicle monitoring, and customer information management. Traditional methods often involve manual

documentation using paper records, notebooks, and spreadsheets. While these methods may be effective for small-scale operations, they become increasingly difficult to manage as the volume of customers and transactions grows. Manual processes can result in inaccurate records, misplaced documents, delayed transaction processing, and difficulties in tracking vehicle availability. These operational issues can negatively affect business performance and reduce customer satisfaction.

One of the most common problems encountered by rental businesses is the occurrence of reservation conflicts. Reservation conflicts arise when multiple customers are assigned to the same vehicle due to inaccurate record keeping or delays in updating vehicle status. Such situations create inconvenience for customers and may damage the reputation of the business. Additionally, manual reservation processes often require employees to spend significant amounts of time verifying bookings, updating records, and communicating with customers. These repetitive tasks increase workload and reduce overall productivity. Implementing an automated reservation system can help eliminate these challenges by providing real-time updates and centralized management of booking information.

Customer satisfaction is considered a critical factor in determining the success of service-oriented businesses. Customers expect rental services to be efficient, reliable, and easily accessible. Delays in processing reservations, inaccurate information, and poor communication can negatively influence customer perceptions and discourage repeat business. Modern customers increasingly prefer online services that allow them to make reservations, review available options, and receive confirmations without visiting a physical office. Therefore, businesses must adopt technological solutions that provide convenience while ensuring accuracy and reliability throughout the rental process.

The emergence of web-based technologies has significantly influenced the development of modern business systems. Web-based applications enable users to access services from virtually any location with an internet connection. Unlike traditional desktop applications, web-based systems provide greater accessibility and flexibility for both customers and administrators. Customers can make reservations, check vehicle availability, and view transaction details using computers, tablets, or smartphones. Meanwhile, administrators can monitor

business operations, manage records, and generate reports from a centralized platform. These capabilities contribute to improved efficiency and better overall service delivery.

Database management is another important aspect of modern information systems. A database serves as a centralized repository for storing, organizing, and retrieving information. In a car rental environment, databases can store customer information, vehicle records, reservation details, payment transactions, and operational reports. The use of a centralized database improves data accuracy by reducing duplication and ensuring consistency across different system functions. Furthermore, database systems facilitate faster retrieval of information, enabling employees to respond more effectively to customer inquiries and operational requirements.

Financial management is essential to the sustainability and growth of any business organization. Accurate recording and monitoring of financial transactions allow business owners to evaluate performance, identify trends, and make informed decisions. In many small rental businesses, financial records are still maintained manually, increasing the risk of calculation errors, missing receipts, and inaccurate reporting. Such issues can affect profitability and hinder effective decision-making. The integration of a cash register system into rental operations provides an efficient solution for managing financial transactions and ensuring accurate documentation of payments and revenues.

A cash register system automates the process of recording transactions, generating receipts, and maintaining financial records. Through automation, businesses can minimize human errors associated with manual computations and record keeping. Cash register systems also support transparency by providing detailed transaction histories that can be reviewed when necessary. In addition, these systems simplify report generation, enabling business owners to monitor daily, weekly, monthly, and annual financial performance. The availability of accurate financial information contributes to better planning and resource allocation within the organization.

The integration of a reservation management system with a cash register system provides numerous benefits for rental businesses. By combining these functions into a single platform, organizations can streamline operations and eliminate redundant processes. Reservation details can be automatically linked to payment records, reducing the need for duplicate data entry. This integration improves accuracy, enhances efficiency, and supports

effective monitoring of business activities. Furthermore, it provides administrators with a comprehensive view of operational and financial information, enabling more informed decision-making.

As technological advancements continue to shape modern business environments, organizations are increasingly investing in digital transformation initiatives. Digital transformation involves the adoption of technologies that improve organizational processes, enhance customer experiences, and create new opportunities for growth. For car rental businesses, digital transformation may include online reservation systems, automated payment processing, vehicle tracking technologies, and data analytics tools. These innovations contribute to greater efficiency and competitiveness while addressing many of the limitations associated with traditional business practices.

The development of a Car Rental with Cash Register System represents an effort to address the operational challenges commonly experienced in vehicle rental businesses. By integrating reservation management and transaction processing into a unified web-based platform, the proposed system aims to improve efficiency, accuracy, and accessibility. The system seeks to provide customers with a convenient method of reserving vehicles while enabling administrators to manage bookings, monitor transactions, and generate reports more effectively. Through automation and centralized data management, the proposed system can support improved business performance and enhanced customer satisfaction.

In the long term, the adoption of automated rental management systems can contribute to the sustainability and growth of transportation service providers. Businesses that utilize technology effectively are better equipped to respond to changing customer demands and market conditions. Automated systems improve operational consistency, reduce administrative burdens, and provide valuable information for strategic planning. As the transportation industry continues to evolve, organizations that embrace technological innovation will be better positioned to achieve operational excellence and maintain a competitive advantage in the marketplace.

In response to these identified challenges, this study proposes the development of "Calma Transportation Services: Implementation of a Cash Register and Reservation System." The main objective of the proposed system is to replace existing manual processes with a fully computerized and automated solution that can

streamline operations and improve overall service delivery. The system is designed to enhance the efficiency of reservation handling, improve the accuracy of payment processing, and ensure reliable data storage through a centralized database system.

By automating key processes such as booking management, payment computation, and record keeping, the system is expected to significantly reduce human errors, minimize operational delays, and improve overall productivity. It will also create a more organized workflow for staff members, allowing them to focus more on customer service and business improvement rather than time-consuming manual tasks.

Ultimately, the implementation of the proposed system will not only benefit the internal operations of Calma Transportation Services but will also greatly enhance the overall customer experience. Faster transaction processing, accurate booking records, improved system reliability, and better service coordination will contribute to higher customer satisfaction. In the long term, this digital transformation will support business growth, increase competitiveness, and ensure that Calma Transportation Services can effectively adapt to the evolving demands of modern transportation systems.

1.2 Statement of the Problem

Calma Transportation Services currently operates using manual methods in managing its vehicle reservations, customer information, and payment transactions. These processes rely heavily on handwritten logbooks, paper-based records, and manual encoding of transactions. While this traditional approach may have been sufficient during the early stages of operation when transaction volume was still limited, it has gradually become inefficient and increasingly difficult to manage as the number of customers and daily transactions continues to grow. The lack of automation in these processes limits the ability of the business to respond quickly and accurately to customer demands.

The continued reliance on manual systems presents several operational challenges that directly affect the efficiency, accuracy, and reliability of business operations. One of the primary concerns is the difficulty in

monitoring vehicle availability in real time. Since records are updated manually, there is a higher possibility of outdated, incomplete, or inconsistent information being recorded. This can lead to scheduling conflicts, overlapping reservations, or double bookings, where more than one customer is assigned to the same vehicle at the same time. These issues not only disrupt daily operations but also negatively affect customer satisfaction, trust, and the overall reputation of the business. In addition, manual systems make it difficult to manage and retrieve important business information. As transaction volumes increase, logbooks, receipts, and handwritten documents become harder to manage and more time-consuming to search through. Staff members may experience delays in retrieving booking histories, customer details, or payment records, especially when dealing with multiple transactions at once. This lack of efficient data organization can significantly slow down daily operations and reduce overall productivity and service efficiency.

Since all calculations are performed by hand, there is a higher risk of human error, especially when handling multiple transactions, discounts, additional fees, or varying rental durations. Even minor mistakes in computation can result in incorrect billing, financial discrepancies, and potential disputes between the business and its customers. Over time, these errors can affect the financial integrity of the business and reduce customer confidence in the accuracy of the services provided.

Furthermore, the absence of a centralized system makes it challenging to generate accurate, timely, and comprehensive reports that are essential for business monitoring and decision-making. Important data such as daily transactions, customer records, payment histories, and vehicle usage statistics are often scattered across different logbooks and physical files. This fragmented system makes it difficult for management to analyze business performance, identify trends, and evaluate operational efficiency. As a result, the business may struggle to make informed decisions that are necessary for improvement and growth.

Another concern is the lack of data security and backup in manual systems. Physical records are vulnerable to damage caused by environmental factors such as fire, water, wear and tear, or accidental misplacement. Once records are lost or damaged, recovering accurate information becomes extremely difficult or

impossible. This poses a serious risk to the continuity and reliability of business operations, especially in cases where historical data is needed for verification or reporting purposes.

Given these challenges, there is a clear and growing need for a computerized reservation and cash register system that can enhance the efficiency, accuracy, and reliability of Calma Transportation Services. The development of such a system is expected to streamline business operations by automating key processes such as booking management, payment processing, and record keeping. It will also improve data organization by providing a centralized database that allows for real-time updates and easier retrieval of information.

Moreover, implementing a digital system will significantly reduce human error, improve transaction speed, and enhance overall service quality. Staff members will be able to perform their tasks more efficiently, while customers will benefit from faster and more accurate service delivery. Ultimately, the transition from a manual system to a computerized platform will not only modernize the operations of Calma Transportation Services but will also support its long-term growth, competitiveness, and ability to meet the increasing demands of the transportation industry.

Specifically, this study seeks to answer the following problems:

1. How can vehicle availability be monitored efficiently in real time to prevent double bookings and scheduling conflicts?
2. How can customer booking records be securely stored, organized, and easily retrieved when needed?
3. How can payment computations and cash register processes be automated to reduce errors in transactions?
4. How can the generation of reports and tracking of transaction history be improved to support better business monitoring and decision-making?

1.3 Objectives of the Study

1.3.1 General Objective

The main objective of this study is to design and develop a web-based Cash Register and Reservation System for Calma Transportation Services that will improve the efficiency, accuracy, reliability, and overall effectiveness of managing vehicle reservations, customer records, and payment transactions through a centralized and automated digital platform. The proposed system aims to replace the existing manual processes, which rely heavily on handwritten records and paper-based documentation, with a computerized solution that streamlines business operations and minimizes the occurrence of human errors.

Specifically, the system seeks to automate key operational activities such as reservation management, vehicle availability monitoring, customer information management, payment processing, and report generation. By integrating these functions into a single platform, the system will provide a more organized and efficient approach to handling daily business transactions. It is also intended to improve the accessibility, security, and accuracy of business records by storing information in a centralized database that can be easily updated, retrieved, and monitored when needed.

Furthermore, the proposed system aims to support better decision-making by providing accurate and timely information regarding reservations, vehicle utilization, customer transactions, and financial records. Through automation and digital record management, the system is expected to reduce administrative workload, enhance productivity, improve service quality, and contribute to the continuous growth and modernization of Calma Transportation Services. Ultimately, the development of the Cash Register and Reservation System seeks to provide a reliable technological solution that will strengthen operational performance, increase customer satisfaction, and support the long-term success of the business.

1.3.2 Specific Objectives

This study aims to achieve the following specific objectives:

- To design and develop a user-friendly system that simplifies and improves the vehicle reservation process, allowing customers and staff to manage bookings more efficiently and accurately.
- To automate the recording, updating, and management of customer booking information in order to reduce manual encoding errors and improve data consistency.
- To implement a real-time monitoring feature that allows users and administrators to track vehicle availability and status, helping to prevent scheduling conflicts and double bookings.
- To develop an automated cash register function that accurately computes rental fees and other related charges, minimizing human errors in financial transactions.
- To create a centralized and secure database that stores all booking, customer, and transaction records in an organized manner for easy retrieval and management.
- To design and implement a reporting feature that generates summaries of reservations and financial transactions, supporting business monitoring and decision-making processes.
- To improve the overall efficiency, reliability, and performance of Calma Transportation Services by integrating all processes into a unified automated system.

1.4 Scope and Delimitation

This study focuses on the design and development of a **web-based Cash Register and Reservation System for Calma Transportation Services**. The primary purpose of the system is to improve the efficiency, accuracy, and reliability of managing vehicle reservations, customer records, and financial transactions by transitioning from a manual and paper-based process to a computerized and automated platform. The proposed system aims to modernize the current operational workflow of the business by providing a more structured, organized, and efficient approach to handling daily transactions and service requests.

The scope of the system includes essential functionalities that are necessary for the smooth operation of a car rental business. These functionalities include vehicle management, customer booking and reservation recording, automated payment processing, and centralized storage of all transaction-related data. Through the system, administrators and staff will be able to add, update, and manage vehicle information, ensuring that records are always accurate and up to date. The reservation module will allow for efficient scheduling of vehicle rentals, reducing the possibility of conflicts and ensuring that all bookings are properly recorded and monitored.

In addition, the system will include a real-time vehicle availability monitoring feature, which provides updated information on whether a vehicle is available, reserved, or currently in use. This feature is essential in preventing double bookings and ensuring that customers receive accurate information regarding available transportation options. The system will also support automated computation of rental fees and charges, ensuring that all financial transactions are processed accurately and consistently without the need for manual calculations.

Furthermore, the system will be capable of generating basic reports related to reservations, customer transactions, and financial summaries. These reports will assist the business in monitoring daily operations, evaluating performance, and supporting decision-making processes. By having access to organized and structured data, management can better understand business trends and identify areas that require improvement.

The system will be developed as a **web-based application**, allowing it to be accessed through standard web browsers such as Google Chrome, Microsoft Edge, or other internet-enabled platforms. It can be used on devices such as desktop computers, laptops, and other compatible devices connected to the internet or local network. This web-based approach ensures flexibility and convenience for users, as it eliminates the need for installation on multiple devices and allows centralized access to system functions.

The primary users of the system will be the **business owner and authorized staff members** of Calma Transportation Services. These users are responsible for managing reservations, handling customer information, processing payments, and maintaining vehicle records. The system is designed to assist them in performing their tasks more efficiently by reducing manual workload, minimizing repetitive processes, and decreasing the

likelihood of human errors. As a result, staff members can focus more on customer service and business improvement rather than administrative tasks.

However, despite the advantages of the proposed system, the study also has certain limitations and delimitations. The system will not include advanced functionalities such as GPS tracking of vehicles, integration with external online payment gateways such as GCash, PayPal, or credit card processing systems, or the development of a dedicated mobile application. These features are excluded due to scope limitations and the focus on core reservation and cash register functionalities.

Additionally, the system will not support full-scale enterprise deployment and will be developed primarily as an academic prototype intended for research, demonstration, and evaluation purposes. As such, it is designed to simulate real-world operations but may still require further enhancement, optimization, and testing before being implemented in a full commercial environment. Future improvements may include system scalability, enhanced security features, and integration with advanced technologies depending on business needs and further development opportunities.

1.5 Significance of the Study

This study is expected to provide meaningful contributions and benefits to various stakeholders, particularly in improving the efficiency and effectiveness of transportation and reservation management processes within Calma Transportation Services. By developing a computerized Cash Register and Reservation System, the study aims to address existing operational challenges and enhance overall service delivery.

Calma Transportation Services

The primary beneficiary of this study is Calma Transportation Services. The proposed system will significantly improve operational efficiency by automating key business processes such as vehicle reservations, customer record management, and payment transactions. This reduces reliance on manual procedures, minimizes

human errors, and ensures more accurate and organized data handling. As a result, the business will be able to provide faster services, improve workflow management, and enhance overall productivity and profitability.

Employees / Staff

The employees or staff members of Calma Transportation Services will benefit from the system through simplified and more efficient work processes. The system provides a centralized and user-friendly platform that allows staff to easily manage reservations, monitor vehicle availability, and process payments. This reduces workload, eliminates repetitive manual tasks, and helps employees focus more on customer service and operational improvement rather than administrative difficulties.

Customers

Customers will greatly benefit from the improved speed, accuracy, and reliability of services. With the implementation of the system, booking transactions become faster and more convenient, reducing waiting time and eliminating common issues such as double bookings or incorrect reservation details. This leads to a more satisfying customer experience and improves trust and confidence in the services provided by Calma Transportation Services.

Researchers / Developers

For the researchers and developers, this study serves as an opportunity to enhance their technical knowledge and practical skills in system development, programming, database design, and system integration. It also allows them to apply theoretical concepts learned in academic settings to a real-world application. Through this process, the researchers gain valuable experience in developing a functional and structured information system.

Future Researchers

This study will serve as a useful reference for future researchers who intend to explore similar topics related to reservation systems, transportation management systems, and business process automation. It may provide insights into system design, development methodologies, and implementation strategies that can be adapted or improved in future studies. Additionally, it can serve as a baseline for further enhancements or more advanced system development projects.

1.6 Definition of Terms

To provide clarity and a better understanding of the key concepts used in this study, the following terms are defined as used in the context of the system:

Booking – The act of reserving a vehicle for use at a specific date and time through the system. It represents a confirmed request made by a customer to secure transportation services. In this study, booking ensures that each reservation is properly recorded, scheduled, and monitored to avoid conflicts such as double bookings or overlapping schedules.

Cash Register – A digital feature of the system that automatically records, computes, and manages payment transactions related to vehicle rentals. It ensures accurate and efficient handling of financial records by reducing manual computation errors and providing real-time transaction summaries for the business.

Customer – An individual who rents or reserves a vehicle from Calma Transportation Services through the reservation system. Customers may include tourists, local residents, or business clients who require transportation services for a specific period of time.

Database – A structured and centralized electronic system used for storing, organizing, managing, and retrieving all relevant data such as customer information, reservations, vehicle details, and financial transactions. It ensures data integrity, consistency, and security within the system.

Reservation – An advance arrangement made by a customer through the system to secure a vehicle for a specific date and time of use. Reservations help ensure that vehicles are properly scheduled and allocated based on customer demand and availability.

Transaction – Any activity recorded within the system that involves booking, reservation, payment, or other service-related processes. Transactions serve as official records of business operations and are essential for monitoring financial performance.

Vehicle Availability – The current status of a vehicle indicating whether it is available for use, reserved, or currently rented. This feature is important in managing fleet operations and preventing scheduling conflicts within the system.

Web-Based System – An application that operates through a web browser and can be accessed via the internet or local network. It allows users such as staff and administrators to manage reservations, payments, and records efficiently without requiring installation on a specific device.

Automation – The use of technology to perform processes automatically with minimal human intervention. In this study, automation is applied in booking management, payment computation, and record handling to improve efficiency and reduce human errors.

Calma Transportation Services – The business organization where the proposed system will be implemented. It is a transportation service provider that offers vehicle rental and reservation services to customers and currently operates using manual processes.

User Interface (UI) – The visual layout and interactive components of the system that allow users to navigate and perform tasks such as booking, payment processing, and record management. A well-designed UI improves usability and enhances user experience.

System – Refers to the proposed Calma Transportation Services: Implementation of a Cash Register and Reservation System, which is a web-based application designed to automate and improve the management of bookings, payments, and transportation operations.

Real-Time Processing – The ability of the system to update and display information instantly as changes occur, such as updating vehicle availability after a booking or payment transaction.

Record Management – The process of organizing, storing, and retrieving data within the system to ensure that all customer, booking, and financial records are properly maintained and easily accessible.

Efficiency – Refers to the ability of the system to perform tasks quickly and accurately while minimizing time, effort, and resource usage in managing reservations and transactions.

Accuracy – The degree to which the system produces correct and error-free results, particularly in booking schedules, payment computations, and record-keeping.

CHAPTER 2

REVIEW OF RELATED LITERATURE AND STUDIES

2.1 Foreign Studies

The rapid advancement of information technology has significantly transformed the way services are delivered and managed across various industries worldwide, particularly in transportation and vehicle rental services. As businesses continue to adopt digital solutions, traditional manual processes are gradually being replaced by web-based systems that offer faster, more accurate, and more efficient methods of handling transactions and operations. This shift toward digitalization is driven by the growing demand for convenience, real-time access to information, and improved service quality.

In the context of vehicle rental services, foreign studies have shown that online car rental systems play a vital role in enhancing operational efficiency by automating key processes such as reservation handling, vehicle availability tracking, payment processing, and customer management. These systems reduce the reliance on manual documentation and face-to-face transactions, which are often prone to delays, errors, and inconsistencies. Instead, web-based platforms allow users to access services anytime and anywhere, improving accessibility and overall user experience.

Moreover, foreign literature highlights that the integration of modern technologies such as real-time databases, mobile applications, automated notifications, and intelligent system features contributes significantly to improving customer satisfaction and business performance. These systems not only streamline internal operations but also promote transparency, reliability, and better communication between service providers and customers.

Overall, foreign studies consistently emphasize that the adoption of web-based car rental systems is essential in modernizing transportation services. They demonstrate that digital transformation leads to improved efficiency, reduced operational costs, and enhanced service delivery, making it a critical component in the development of modern car rental management systems.

Osman et al. [1] developed an online car rental system that combines web-based technology with SMS notifications to support communication between rental companies and customers. The study focused on making vehicle reservations more convenient and providing customers with updates about their rental requests. Using an online platform and SMS technology helps make reservation information easier to access and allows customers to receive important notifications without always contacting the rental office.

The system allows customers to make car rental reservations online and uses SMS technology to communicate reservation-related information. By combining the website with text message notifications, the system aims to improve the handling of rental requests and make communication between customers and rental staff more convenient.

The study demonstrates how integrating SMS notifications into a web-based car rental system can support reservation management and customer communication. This approach can help rental businesses organize their services and provide customers with a more convenient way to receive rental updates.

This study is relevant to the proposed **Calma Transportation Services: Implementation of a Cash Register and Reservation System** because both systems use digital technology to improve car rental operations. While Osman et al. [1] focus on online reservations and SMS communication, the proposed system also includes cash register transactions, payment records, automated billing, and sales reports. These features aim to improve reservation handling, maintain accurate financial records, and make daily rental operations more efficient [1].

Azmi and Hassan [2] developed a web-based automobile rental system to improve the management and accessibility of vehicle rental services. The study focused on using digital technology to make rental activities more organized and convenient for customers. Compared with traditional rental procedures, a web-based system can help businesses manage vehicle information and reservation records in a more structured way.

The system provides an online platform for handling automobile rental information and supporting rental-related activities. By organizing information through a digital system, rental businesses can manage their records more efficiently and make important vehicle and reservation details easier to access. This approach also supports better coordination between customers and rental staff.

The study highlights the potential of web-based technology to improve the organization and management of automobile rental services. Using a digital platform can reduce dependence on paper-based records and help make rental information easier to maintain and retrieve.

This study is relevant to the proposed **Calma Transportation Services: Implementation of a Cash Register and Reservation System** because both projects aim to improve rental operations through digital technology. While Azmi and Hassan [2] focus on automobile rental management, the proposed system also includes cash register transactions, payment records, automated billing, and sales reports. These additional features aim to improve financial recordkeeping and support more efficient reservation and rental management [2].

Harissa [3] developed a web-based car rental information system using the Agile Scrum methodology to improve the management of rental services. The study focused on developing a system that could meet user requirements through an organized and flexible development process. Agile Scrum allows developers to divide the work into smaller tasks, review progress regularly, and make improvements based on testing and feedback.

The researcher designed and implemented a web-based prototype for managing car rental information. The development process followed the Scrum approach, allowing the system to be developed and reviewed in

stages. This approach supports the identification of problems during development and helps ensure that the system's features align with the intended requirements.

The study demonstrates how Agile Scrum can support the structured development of a web-based car rental system. By developing and reviewing the system incrementally, developers can identify issues earlier and make necessary adjustments before completing the project.

This study is relevant to the proposed **Calma Transportation Services: Implementation of a Cash Register and Reservation System** because both projects involve developing a web-based system to improve car rental management. Harissa [3] provides a reference for using Agile Scrum in system development, while the proposed system focuses on reservation management, cash register transactions, payment records, and sales reports. These features aim to make rental operations more organized and improve the accuracy of transaction records [3].

Yang et al. [4] investigated methods for estimating car rental demand to help improve fleet planning and resource allocation. The researchers addressed the challenge of estimating demand when different types of rental vehicles may be needed at different times. Accurate demand estimates are important because rental companies need to understand customer needs and manage their available vehicles effectively.

The study proposed a method for estimating demand across multiple vehicle types without relying on restrictive assumptions about the distribution of rental demand. By analyzing demand-related data, the approach supports a better understanding of how rental needs vary among vehicle categories. This information can help rental businesses make more informed decisions about fleet allocation and vehicle availability.

The study highlights the importance of data analysis in understanding rental demand and supporting operational planning. Reliable demand estimates can provide useful information for managing vehicle resources and preparing for changes in customer needs.

This study is relevant to the proposed **Calma Transportation Services: Implementation of a Cash Register and Reservation System** because both projects emphasize the value of organized data in managing car

rental operations. While Yang et al. [4] focus on estimating demand for different vehicle types, the proposed system focuses on reservation management, cash register transactions, payment records, and sales reports. The records generated by the proposed system may also help the business review rental activity and make better operational decisions [4].

Caballero-Gil et al. [5] developed a decentralized car rental system using Ethereum blockchain technology to improve the transparency and reliability of rental transactions. The researchers explored how blockchain could support rental operations by recording transaction information in a distributed ledger. Unlike traditional systems that depend on a centralized database, blockchain allows recorded transactions to be verified and traced.

The system uses smart contracts to manage rental activities according to predefined conditions. These contracts can automatically execute specific actions when the required conditions are met, helping reduce the need for manual processing. The use of blockchain also provides a verifiable record of transactions, supporting greater transparency between customers and rental providers.

The study highlights the potential of blockchain technology to automate rental processes and improve transaction transparency. However, the suitability of this technology depends on the needs and resources of the business implementing it.

This study is relevant to the proposed **Calma Transportation Services: Implementation of a Cash Register and Reservation System** because both projects aim to improve rental transaction management and recordkeeping. Although Caballero-Gil et al. [5] focus on blockchain and smart contracts, the proposed system focuses on reservation management, cash register transactions, payment records, and sales reports. These features are intended to maintain organized financial records and support more reliable rental operations [5].

Ogbiti and Aaron [6] developed a web-based car rental management system to improve the organization and efficiency of rental operations. The study addressed challenges associated with managing rental activities manually, particularly in maintaining customer information, vehicle records, and rental transactions. Using a web-based platform can help businesses organize these records and make rental information easier to access and manage.

The researchers designed a system that supports car rental management through computerized processes. The study focuses on using web technology to organize rental information and support the handling of rental activities. By moving away from manual recordkeeping, the system provides a more structured approach to managing information within a car rental business.

The study highlights the usefulness of web-based systems in organizing rental records and supporting business operations. Centralized information management can help staff retrieve and update records more conveniently, reducing dependence on paper-based procedures.

This study is relevant to the proposed **Calma Transportation Services: Implementation of a Cash Register and Reservation System** because both projects aim to improve car rental operations through digital technology. While Ogbiti and Aaron [6] focus on web-based rental management, the proposed system also integrates cash register transactions, payment records, automated billing, and sales reports. These features aim to improve financial recordkeeping and make reservation and rental transactions easier to manage [6].

Muttaqin, Wadly, and Veranda [7] discussed the development of a web-based car rental information system to support the digital transformation of rental services. The study focused on using web technology to make rental operations more organized and accessible. By moving rental activities to an online platform, businesses can manage important information more conveniently and reduce their dependence on traditional recordkeeping methods.

The system provides a digital platform for managing car rental information and supporting rental-related activities. A web-based approach allows vehicle and reservation information to be organized in one place, making it easier for rental staff to access and maintain records. This can also help improve coordination between customers and the rental business during the reservation process.

The study highlights the role of digital technology in modernizing car rental services and supporting more organized information management. A centralized web-based system can help businesses handle rental information more systematically and provide a convenient way to manage their services.

This study is relevant to the proposed **Calma Transportation Services: Implementation of a Cash Register and Reservation System** because both projects aim to improve car rental operations through digital technology. While Muttaqin et al. [7] focus on a web-based car rental information system, the proposed system also includes cash register transactions, payment records, automated billing, and sales reports. These features aim to improve financial recordkeeping and support more efficient reservation and rental management [7].

Rahman and Rusli [8] developed a web-based car rental business management system to help organize rental activities and improve the management of business information. The study focused on using digital technology to support rental operations and reduce the difficulties associated with traditional management methods. A web-based system provides a more organized way to handle rental information and makes it easier for businesses to maintain important records.

The system was designed to support car rental business management through an online platform. By organizing rental-related information in a digital environment, the system helps make business records easier to access, update, and manage. This approach can support better organization of rental activities and help staff handle information more systematically.

The study highlights the value of web-based technology in improving the management of car rental businesses. Digital information management can help reduce dependence on manual documentation and provide a more structured way to maintain rental records.

This study is relevant to the proposed **Calma Transportation Services: Implementation of a Cash Register and Reservation System** because both projects aim to improve car rental management through digital technology. While Rahman and Rusli [8] focus on web-based business management, the proposed system also includes reservation processing, cash register transactions, payment records, automated billing, and sales reports. These features aim to improve financial recordkeeping and support more organized rental operations [8].

Oktaviani and Sari [9] developed a web-based car rental information system for CV. Rafael Trans to improve the management of vehicle rental activities. The study focused on using digital technology to organize rental information and support the handling of customer reservations. A web-based platform provides a more systematic way to manage vehicle and rental records compared with relying entirely on manual documentation.

The system was designed to support car rental information management through an online platform. By organizing rental-related data in one system, the application helps staff manage vehicle information and reservation records more conveniently. This approach supports better organization of rental information and makes records easier to maintain.

The study highlights the usefulness of web-based information systems in helping car rental businesses manage their operations. Digital recordkeeping can reduce dependence on paper-based processes and provide a more organized way to handle rental information.

This study is relevant to the proposed **Calma Transportation Services: Implementation of a Cash Register and Reservation System** because both projects aim to improve car rental operations through digital technology. While Oktaviani and Sari [9] focus on car rental information management, the proposed system also

includes reservation processing, cash register transactions, payment records, automated billing, and sales reports. These features aim to improve financial recordkeeping and support more efficient rental operations [9].

Karsa et al. [10] conducted a study on the development of a web-based car rental information system for CV. Mandarental Cirebon. The study focused on using web technology to improve the management of car rental services and organize rental-related information. By introducing a digital platform, the researchers aimed to provide a more systematic approach to handling vehicle details and customer reservations.

The developed system was designed to support the management of car rental information through a web-based platform. This type of system can help organize vehicle records, customer details, and reservation information in one place. Centralizing these records also provides a more structured way for rental businesses to access and manage important information related to their daily operations.

The study highlights the potential of web-based information systems in supporting car rental management. Digital platforms can help organize rental records and make information easier to manage compared with relying entirely on manual documentation. However, specific claims about reduced processing time, fewer errors, or real-time vehicle updates should be included only if they are confirmed by the original study.

This study is relevant to the proposed **Calma Transportation Services: Implementation of a Cash Register and Reservation System** because both projects aim to improve car rental operations through digital technology. While Karsa et al [10]. focused on a web-based car rental information system, the proposed system also includes cash register functions for recording payments, managing sales transactions, and generating financial reports. These additional functions aim to support organized reservation management and accurate financial record-keeping for Calma Transportation Services [10].

Sari and Jihad [11] conducted a study on the design of a web-based car rental service portal information system for 123 Lampung Utara. The study focused on using web technology to support the management of car rental services and organize rental-related information. By developing an online portal, the researchers aimed to provide a more systematic way of managing vehicle rental information and customer bookings.

The proposed system was designed to organize car rental information through a web-based platform. This type of platform can help manage vehicle details, customer records, and reservation information in one place. By storing rental-related information digitally, the system provides a more structured approach to accessing and managing records compared with relying entirely on manual documentation.

The study highlights the use of web-based information systems in managing car rental services. Digital platforms can help organize rental information and make records easier to access and maintain. However, specific claims about improved efficiency, fewer errors, real-time updates, or increased customer satisfaction should only be included when they are confirmed by the original study.

This study is relevant to the proposed **Calma Transportation Services: Implementation of a Cash Register and Reservation System** because both projects aim to improve car rental operations through digital technology. While Sari and Jihad [11] focused on a web-based car rental service portal, the proposed system also includes cash register functions for recording payments, managing sales transactions, and generating financial reports. These additional functions aim to support organized reservation management and accurate financial record-keeping for Calma Transportation Services [11].

Wana, Candra, and Iryani [12] developed a web-based car rental information system for Sasha Car Rental that uses a WhatsApp Gateway to support communication between the rental service and its customers. The study focused on using web technology to organize car rental information and make communication more convenient. By integrating WhatsApp messaging into the system, the researchers explored a digital approach to supporting customer communication during the rental process.

The system combines a web-based platform with WhatsApp messaging to support rental-related communication. This approach can provide customers and rental staff with a more convenient way to exchange information. Integrating messaging into a car rental information system also demonstrates how communication tools can complement digital rental management processes.

The study highlights the potential of combining web-based information systems with messaging technology to support car rental services. This approach can help make communication more accessible and organize rental-related information within a digital platform. However, specific claims about improved efficiency, reduced errors, or increased customer satisfaction should only be included if they are supported by the original study.

This study is relevant to the proposed **Calma Transportation Services: Implementation of a Cash Register and Reservation System** because both projects aim to improve car rental operations through digital technology. While Wana et al. [12] emphasized web-based rental information and WhatsApp communication, the proposed system also focuses on reservation management, cash register transactions, payment records, and sales reports. These features aim to support more organized reservations and accurate financial record-keeping for Calma Transportation Services [12].

Ahmed and Yunus [13] developed the AL-WEFAQ Car Rental Reservation System to support the management of car rental services through a digital platform. The study focused on providing an online reservation system that helps organize vehicle rental information and customer bookings. By using web technology, the researchers aimed to provide a more accessible and systematic way of handling car rental reservations.

The system focuses on managing car rental reservations and related information through an online platform. A web-based reservation system can help organize vehicle details and booking records in one place, making rental information easier for customers and rental staff to access. This approach also demonstrates how

digital technology can support the transition from traditional rental procedures to a more organized reservation process.

The study highlights the role of web-based reservation systems in supporting car rental operations. By organizing reservation information digitally, rental businesses can manage bookings and related records more systematically. However, specific claims about significant improvements in efficiency, reduced errors, or increased customer satisfaction should only be included if these findings are confirmed by the original study.

This study is relevant to the proposed **Calma Transportation Services: Implementation of a Cash Register and Reservation System** because both projects focus on improving car rental services through digital technology. While Ahmed and Yunus [13] focused on the AL-WEFAQ reservation system, the proposed system also includes cash register functions for recording payments, managing sales transactions, and generating financial reports. These additional functions aim to support organized reservation management and accurate financial record-keeping for Calma Transportation Services [13].

Jadhav and Karande [14] developed an online car rental system to support vehicle rental services through a web-based platform. The study focused on using digital technology to make rental information and reservation services more accessible to customers. By providing an online platform, the system demonstrates an approach to organizing rental activities and simplifying the process of accessing car rental services.

The system focuses on providing online access to car rental information and reservation-related activities. A web-based platform can help customers access rental details more conveniently while allowing rental staff to organize vehicle information and booking records. This approach also shows how digital systems can help reduce dependence on manual procedures in managing car rental services.

The study highlights the potential of online car rental systems to support more organized rental operations. By placing rental information and reservation activities on a digital platform, businesses can manage records in a

more systematic way. However, specific claims about reduced booking time, fewer errors, real-time vehicle availability, or increased customer satisfaction should only be included if they are confirmed by the original study.

This study is relevant to the proposed **Calma Transportation Services: Implementation of a Cash Register and Reservation System** because both projects aim to improve car rental operations through digital technology. While Jadhav and Karande [14] focused on an online car rental system, the proposed system also includes cash register functions for recording payments, managing sales transactions, and generating financial reports. These additional functions aim to support organized reservation management and accurate financial record-keeping for Calma Transportation Services [14].

Bilal and Hanif [15] developed a web-based vehicle rental application for PT. Duta Rental Mobil using the Waterfall software development methodology. The study focused on improving the management of vehicle rental services by replacing manual processes with a computerized system. The researchers addressed challenges in managing rental records, vehicle information, and transaction reports, which can become difficult to maintain when handled manually.

To address these challenges, the researchers developed a web-based application using PHP and MySQL. The system includes functions for customer registration, login, vehicle selection, online booking, payment processing, and rental history. It also provides administrative features for managing customer information, vehicle records, reservations, and transactions. By storing rental information in a centralized database, the system provides a more organized way to manage rental activities.

The study reported that the developed application was evaluated through alpha and beta testing, with beta testing achieving an 88% user satisfaction score. This result indicates a positive evaluation from users based on the study's assessment. The researchers also identified GPS-based vehicle tracking as a possible future enhancement to expand the system's functionality.

This study is relevant to the proposed **Calma Transportation Services: Implementation of a Cash Register and Reservation System** because both projects aim to improve vehicle rental operations through a web-based information system. While Bilal and Hanif [15] focused on managing vehicle rentals, customer information, and payment processes, the proposed system also emphasizes cash register transactions, sales history, and financial reports. These functions aim to support more organized reservation management and accurate transaction records for Calma Transportation Services [15].

Bakar and Yuspani [16] conducted a study titled "Initial Stage in Developing an Online Car Rental System (OCRS) Using Customization Business Model." Their research focused on the initial development of an online car rental system to support the delivery and management of vehicle rental services. The study explored how an online platform could help organize rental activities while considering the specific needs and requirements of the business.

The researchers focused on the early stages of developing an Online Car Rental System (OCRS) using a customization business model. This approach emphasizes considering business requirements when planning a system and identifying features that suit the intended users. The study provides a foundation for understanding how an online car rental platform can be planned according to the operational needs of a rental business.

The study highlights the importance of considering business requirements during the initial stages of system development. By identifying the needs of the business before implementation, developers can plan system features that align with rental operations and customer requirements. Since the research focuses on the initial development stage, specific claims about improved efficiency, reduced errors, or increased customer satisfaction should not be made unless supported by the study's findings.

This study is relevant to the proposed **Calma Transportation Services: Implementation of a Cash Register and Reservation System** because both projects focus on using technology to support car rental services. While Bakar and Yuspani [16] focused on the initial development of an online car rental system, the proposed

Calma system also includes cash register functions, payment recording, reservation management, and transaction history. These additional features aim to help the business organize rental operations and maintain accurate customer and financial records [16].

Padhiyar and Rathod [17] presented a web-based car management and rental system designed to support the organization and management of vehicle rental services. Their study focused on using web technology to make rental information more accessible and provide a digital approach to managing car rental activities. The study demonstrates how a website-based system can support the management of vehicle information and rental services.

The researchers explored a web-based approach to car rental management. This type of system can provide an online platform for presenting vehicle information and rental services to customers. It can also help rental businesses organize vehicle records and manage rental-related information within a digital system, reducing their dependence on manual documentation.

The study highlights the potential of web-based technology in supporting car rental management and improving access to rental information. Organizing vehicle and rental records digitally can help businesses maintain information in a more systematic manner. However, specific claims about improved efficiency, reduced errors, or increased customer satisfaction should only be included if they are supported by the original study.

This study is relevant to the proposed **Calma Transportation Services: Implementation of a Cash Register and Reservation System** because both projects use web technology to support vehicle rental operations. While Padhiyar and Rathod [17] focused on car management and rental services, the proposed Calma system also includes reservation management, payment recording, cash register functions, and transaction history. These additional features aim to support organized bookings and accurate financial record-keeping for Calma Transportation Services [17].

Gunasekaran and Arbain [18] presented DriveEasy, a car rental management system designed to support the organization of vehicle rental activities through a digital platform. Their study focused on using technology to improve the management of rental information and provide a more systematic approach to car rental operations. The system demonstrates how digital solutions can be applied to support the needs of rental businesses and their customers.

The DriveEasy system highlights the use of technology in organizing car rental information and related activities. A digital platform can help rental businesses manage vehicle records, organize reservation information, and make rental details easier to access. Using a structured development process can also help ensure that the system is planned and developed according to the intended requirements of the business.

The study emphasizes the role of digital systems in supporting the management of car rental services. Organizing rental information electronically can provide a more systematic way to maintain records and handle daily activities. However, specific claims about improved efficiency, reduced errors, or increased customer satisfaction should only be included if they are supported by the original study.

This study is relevant to the proposed **Calma Transportation Services: Implementation of a Cash Register and Reservation System because** both projects aim to improve car rental management through technology. While DriveEasy focuses on car rental management, the proposed Calma system also includes cash register transactions, payment records, reservation management, and sales reports. These additional features aim to support accurate financial recording and more organized rental operations for Calma Transportation Services [18].

Soares, da Costa, and Bernardo [19] developed a web-based car rental management application for Kailelo Car Rental using PHP and MySQL. The study focused on improving the management of vehicle rental activities through a computerized system. By replacing manual procedures with a digital platform, the researchers aimed to organize vehicle information, customer records, and rental transactions more systematically.

The researchers used the Waterfall software development methodology to develop the application. PHP was used for programming, while MySQL served as the database management system. UML diagrams and an Entity Relationship Diagram (ERD) were used during system design. The application includes administrative modules for dashboard monitoring, brand management, vehicle information, customer records, destination management, order processing, account management, and report generation for customers, rentals, and vehicles.

The study reported that the developed application supported the management of rental information and business activities. Its modules were reported to function properly, allowing the company to organize records and manage rental operations through a centralized platform. These findings demonstrate how a web-based application can support the digital management of vehicle rental services.

This study is relevant to the proposed **Calma Transportation Services: Implementation of a Cash Register and Reservation System** because both projects use web technology and a centralized database to manage rental operations. While Soares et al. [19] focused on vehicle, customer, order, and report management, the proposed Calma system also includes cash register transactions, payment records, sales history, and financial reports. These additional features aim to support organized reservation management and accurate financial record-keeping for Calma Transportation Services [19].

Waje et al. [20] conducted a study titled “Car Rental System” that focused on using technology to support the management of vehicle rental services. Their research explored how a computerized system could help organize rental activities and customer information. By using a digital approach, the study aimed to provide a more systematic way of managing car rental operations and maintaining rental-related records.

The researchers presented a car rental system designed to support the management of vehicle and rental information. A computerized platform can help organize customer records, vehicle details, reservations, and transaction information in one system. This approach can make rental records easier to maintain and provide a more structured way for rental businesses to manage their daily activities.

The study highlights the potential of computerized systems in organizing car rental operations and managing rental information. By reducing dependence on manual documentation, digital systems can help businesses maintain more organized records and handle rental activities systematically. However, specific claims about significant efficiency improvements, real-time vehicle availability, reduced booking errors, or increased customer satisfaction should only be included if they are supported by the original study.

This study is relevant to the proposed **Calma Transportation Services: Implementation of a Cash Register and Reservation System** because both projects focus on using technology to support car rental operations. While Waje et al. [20] focused on a car rental system, the proposed Calma system also includes cash register transactions, payment records, reservation management, sales history, and financial reports. These additional functions aim to support organized rental operations and accurate financial record-keeping for Calma Transportation Services [20].

Nahnisha et al. [21] conducted a study titled “Online Car Rental System” that focused on using web technology to improve the management and accessibility of vehicle rental services. The study explored how an online platform could provide customers with a more convenient way to access rental information and make reservations. By using a digital system, the researchers aimed to support a more organized approach to handling car rental activities.

The researchers developed a web-based car rental system supported by a MySQL database and the XAMPP server environment. The system focused on managing rental information and supporting online reservation activities. A web-based platform can help organize customer details, vehicle information, and booking records, allowing rental businesses to maintain their information in a more systematic way.

The study highlights the potential of online car rental systems to make rental information more accessible and support the digital management of reservations. Using a database to store rental-related information can help businesses organize customer and vehicle records more conveniently. However, specific claims about reduced

processing time, fewer errors, improved productivity, or increased customer satisfaction should only be included if they are supported by the original study.

This study is relevant to the proposed **Calma Transportation Services: Implementation of a Cash Register and Reservation System** because both projects use web technology to improve car rental operations. While Nahnisha et al. [21] focused on online vehicle rental and reservation management, the proposed Calma system also includes cash register transactions, payment records, sales history, and financial reports. These additional features aim to support organized reservations and accurate financial record-keeping for Calma Transportation Services [21].

Philogin and Prasath [22] presented an online car rental administration system designed to support the management of vehicle rental services through digital technology. Their study focused on organizing rental activities and making vehicle information more accessible to users. The system demonstrates how an online platform can support the management of car rental services and provide customers with a more convenient way to explore rental options.

The system includes features for managing reservations, customer information, and rental history. It also provides filtering options that allow users to search for vehicles based on criteria such as vehicle category, price, and availability. These features help organize vehicle information and support the process of selecting suitable rental options through an online platform.

The study highlights how digital tools can support vehicle selection and the organization of rental activities. Features such as vehicle filtering and rental history management can help users access relevant information and allow rental businesses to maintain more organized records. However, specific claims about reduced operating costs, improved resource utilization, or faster decision-making should only be included when supported by the original study.

This study is relevant to the proposed **Calma Transportation Services: Implementation of a Cash Register and Reservation System** because both systems aim to improve rental operations through digital reservation management and organized rental records. While Philogin and Prasath [22] focused on online rental administration and vehicle selection, the proposed Calma system also includes cash register transactions, payment history, automated billing, and sales reports. These additional features aim to support financial monitoring and accurate transaction recording for Calma Transportation Services [22].

Qi, Shibghatullah, and Subaramaniam [23] conducted a study titled "Rehabilitating Flood-Damaged Cars for Sustainable Car Rental Services: A Web-Based System." Their research focused on developing a web-based car rental system that supports sustainable transportation by making properly rehabilitated flood-damaged vehicles available for rental. The study explored how digital technology can support vehicle rental services while promoting the continued use of restored vehicles and reducing unnecessary waste.

The researchers developed the system using the Rapid Application Development (RAD) methodology. The development process focused on identifying system requirements, designing the user interface, constructing the application, and implementing the system. The web-based platform was designed to support access to rental services and the management of vehicle-related information. The researchers also identified GPS tracking and transparent disclosure of vehicle rehabilitation history as possible future enhancements.

The study reported positive results from user acceptability testing, indicating that the system was considered acceptable by its intended users. It also highlighted the importance of providing clear information about vehicle condition and rehabilitation to support customer trust. These findings demonstrate how web-based technology can support sustainable car rental services, although the specific testing results and conclusions should be checked against the original paper before being included in the final documentation.

This study is relevant to the proposed **Calma Transportation Services: Implementation of a Cash Register and Reservation System** because both projects use web technology to support car rental operations.

While Qi et al. [23] focused on sustainable vehicle rental and the management of rehabilitated vehicles, the proposed Calma system focuses on reservation management, cash register transactions, payment records, and sales reports. Both studies demonstrate the potential of digital systems to organize rental information and support more systematic business operations [23].

The reviewed foreign studies highlight how digital technology can support car rental operations. Osman et al. [1] developed a web-based car rental system with SMS notifications to provide customers with rental information and vehicle availability updates. Harissa [3] focused on designing and implementing a web-based car rental information system using Agile Scrum, while Ogbiti and Aaron [6] and Rahman and Rusli [8] examined web-based approaches to car rental management. These studies demonstrate how digital systems can help organize rental information and support reservation and management activities.

Other studies focus on specific technologies and methods that address different needs in the car rental industry. Yang et al. [4] developed a method for estimating and forecasting demand for different vehicle types, reporting an average estimation error of 3.06% in their case study. Caballero-Gil et al. [5] presented a decentralized car rental system using Ethereum smart contracts, while Wana et al. [12] explored a web-based rental system integrated with a WhatsApp gateway. These studies illustrate how demand forecasting, blockchain technology, and communication tools can support particular aspects of car rental services.

Overall, the reviewed literature provides a foundation for developing digital systems that support reservation management, customer communication, vehicle information, and rental operations. These studies are relevant to the proposed ***Calma Transportation Services: Implementation of a Cash Register and Reservation System***, which aims to combine reservation management with cash register transactions, payment records, and sales reports. Unlike systems that primarily focus on reservations, vehicle management, or communication, the proposed system also emphasizes recording financial transactions and organizing sales information. Integrating these functions into one platform aims to help Calma Transportation Services manage bookings and financial records in a more systematic and organized manner.

2.2 Local Studies

The continuous development of information technology in the Philippines has significantly influenced the improvement and modernization of various service-oriented industries, including transportation and vehicle rental services. As the demand for faster, more efficient, and more accessible services continues to grow, many local systems have gradually shifted from traditional manual processes to computerized and web-based platforms. This transition is driven by the need to reduce operational inefficiencies, minimize human errors, and improve the overall quality of service delivery.

In many local settings, especially within small to medium-sized rental businesses, traditional car rental operations still rely on manual record-keeping, walk-in transactions, and basic communication methods such as phone calls or messaging applications. These processes often lead to challenges such as booking conflicts, delayed transaction processing, difficulty in tracking vehicle availability, and inconsistencies in customer records. As a result, service providers experience reduced efficiency, while customers encounter inconvenience and longer waiting times.

To address these issues, several local studies in the Philippines have explored the development and implementation of web-based systems that automate reservation processes, centralize data management, and improve transaction handling. These systems are designed to provide real-time access to information, allowing both administrators and users to efficiently manage bookings, monitor availability, and process transactions more accurately.

Furthermore, local literature emphasizes that the integration of automated systems not only improves operational performance but also enhances customer satisfaction by providing more convenient and accessible services. The use of centralized databases, user-friendly interfaces, and structured system development methodologies has been shown to significantly improve system reliability and efficiency.

Overall, local studies in the Philippines consistently demonstrate that web-based systems are essential in modernizing rental and service-based operations. They highlight the importance of automation and digital transformation in improving efficiency, accuracy, and accessibility, which strongly supports the development of a Car Rental with Cash Register System.

Albino and Acebedo [24] conducted a study titled “Development of Car Rental Management System with Scheduling Algorithm” in 2021. The study focused on developing a car rental management system that uses a scheduling algorithm to help organize vehicle rentals and reservations. It aimed to address challenges in managing vehicle schedules and rental requests.

The researchers developed a system that incorporates a scheduling algorithm to manage rental schedules and vehicle assignments. The system was designed to organize reservation information and help manage vehicle availability. By using a scheduling algorithm, the system aimed to reduce conflicts in vehicle bookings and support more organized rental operations.

The study highlights the importance of scheduling algorithms in managing vehicle availability and rental reservations. This approach can help rental businesses organize booking schedules and identify potential conflicts. However, specific claims about the system’s accuracy, efficiency, and user satisfaction should be based on the actual evaluation results reported in the original study.

This study is relevant to the proposed **Calma Transportation Services: Implementation of a Cash Register and Reservation System** because both projects aim to improve car rental operations through computerized management. While Albino and Acebedo [24] focused on vehicle scheduling and reservation management, the proposed Calma system also includes cash register transactions, payment records, and sales reports. These additional features aim to help the business manage reservations and financial information in one integrated platform [24].

Golo and Encarnacion [25] conducted a study titled “Revolutionizing the Rental Services in Siargao Island: Basis for Developing an Online Vehicle Rental Management System” in 2024. The study focused on the challenges faced by vehicle rental services in Siargao Island and explored the need for an online management system. It aimed to provide a basis for improving rental operations through digital technology.

The researchers examined the existing vehicle rental services to identify areas that could benefit from an online system. The proposed system was intended to support the management of vehicle rental information and reservations. By providing an online platform, the system aimed to make rental information more accessible and help organize rental-related activities.

The study highlights the potential of an online vehicle rental management system to address operational challenges in the rental industry. It provides a basis for using digital technology to organize rental information and support reservation management. However, specific findings about system efficiency, customer satisfaction, or reduced booking errors should be confirmed using the original study.

This study is relevant to the proposed **Calma Transportation Services: Implementation of a Cash Register and Reservation System** because both projects aim to improve vehicle rental operations through digital technology. While Golo and Encarnacion [25] focused on developing an online vehicle rental management system for Siargao Island, the proposed Calma system also includes cash register transactions, payment records, and sales reports. These additional features aim to help the business manage reservations and financial transactions through one integrated platform. [25]

De Jesus et al.[26] developed *PARAM: A Web-based Booking and Management System with Mobile Tracking Application for Param Car Rentals* at Mabalacat City College in Pampanga, Philippines. The study aimed to improve the car rental process by replacing manual booking procedures with a web-based system that helps manage vehicles, reservations, and renters.

The system was designed for three types of users: administrators, renters, and business partners. Its features included vehicle information management, scheduling to check vehicle availability, booking approval or cancellation, renter registration, email validation, and mobile tracking. The researchers used a prototyping methodology to develop the system and tested its functionality, reliability, compatibility, and security.

The researchers reported that the system was successfully implemented and tested according to its specified requirements. They also recommended adding SMS notifications for bookings and extending mobile application support to iOS in future development. These recommendations show that notifications and platform compatibility can be important considerations in improving a car rental system

This study is relevant to the proposed **Calma Transportation Services: Implementation of a Cash Register and Reservation System** because both systems aim to improve vehicle reservation and rental management. However, the proposed Calma system places additional emphasis on cash register transactions, payment records, and sales reports. The PARAM study provides a useful reference for designing reservation workflows and managing vehicle availability, while the Calma system addresses the additional need for organized financial records. [26]

Grepon et al. [27] conducted a study titled “Designing and Implementing e-School Systems: An Information Systems Approach to School Management of a Community College in Northern Mindanao, Philippines.” Published in 2022, the study focused on developing a School Management Information System (SMIS) to address the challenges of handling school records and transactions manually. The researchers aimed to centralize information storage, processing, and retrieval to support the administrative operations of a community college.

The researchers used the Agile Model in developing the system. The e-school system included different modules designed to manage school-related information and transactions. By centralizing records in one system,

the platform was intended to reduce paperwork and make information easier for authorized personnel to access and manage.

The researchers evaluated the system using the ISO 25010 software quality model. The results showed that the system received average ratings above 4.04 for the evaluated criteria, indicating very good performance in functionality, usability, and reliability. The study also reported that respondents believed the system helped improve the school's transaction processes and benefited faculty and staff.

This study is relevant to the proposed **Calma Transportation Services: Implementation of a Cash Register and Reservation System** because both projects aim to organize business or administrative operations through a centralized information system. While Grepon et al. [27] focused on school records and transactions, the proposed Calma system focuses on vehicle reservations, cash register transactions, payment records, and sales reports. Both studies highlight the value of centralized data management in supporting more organized and systematic operations [27].

Marafo et al. [28] conducted a study titled "Arkila: A Progressive Web App Car Rental Management System for Kataguan Rides in La Trinidad, Benguet" in 2025. The study focused on developing a Progressive Web Application (PWA) to address inefficiencies in manual vehicle booking and rental management. It aimed to provide a more organized way to handle reservations and coordinate vehicle availability for a local car rental service.

The researchers used the Feature-Driven Development (FDD) methodology to develop the system through stages involving overall modeling, feature planning, design, development, and deployment. The application used Next.js for the frontend and Firebase and Appwrite for backend services. The researchers conducted unit testing to check system functionality and User Acceptance Testing (UAT) to evaluate the system's functionality and usability.

The results of the UAT produced an overall mean score of 3.72, indicating that users highly accepted the system. The study reported that Arkila achieved its objectives of streamlining booking transactions, improving vehicle coordination, and supporting rental operations. The researchers also recommended future enhancements, including GPS-based fleet tracking and improved offline and mobile functionality.

This study is relevant to the proposed **Calma Transportation Services: Implementation of a Cash Register and Reservation System** because both projects aim to improve car rental operations through digital technology. While Marafo et al. [28] focused on online booking, vehicle coordination, and rental management, the proposed Calma system also emphasizes cash register transactions, payment records, and sales reports. These additional features aim to help Calma Transportation Services manage reservations and financial transactions in one integrated platform [28].

Estoque et al. [29] examined how service quality, assurance, and safety influence customer satisfaction and referrals in car rental services in Malaybalay City, Bukidnon. The study focused on understanding how customers' experiences and perceptions of safety affect their willingness to recommend a rental company to others.

The researchers used a quantitative descriptive-correlational research design and collected survey responses from 287 customers of selected car rental companies. They evaluated different service quality factors, including reliability, responsiveness, assurance, empathy, tangibility, safety, and overall customer satisfaction. The data were analyzed using weighted means and Pearson correlation.

The findings showed that customers generally expressed satisfaction with the service quality and safety of the car rental companies. Assurance, which includes employee competence, courtesy, and trustworthiness, emerged as the strongest predictor of customer referrals. The study also found significant relationships between safety, service quality, and customers' willingness to recommend a rental service.

This study is relevant to the proposed **Calma Transportation Services: Implementation of a Cash Register and Reservation System** because it highlights the importance of customer trust, safety, and service

quality in car rental operations. While Estoque et al.[29] focused on customer perceptions and referrals, the proposed system focuses on improving reservation management, recording payment transactions, and organizing sales records. These features can support more reliable and organized service, contributing to a better customer experience. [29]

Calosa et al. [30] developed Transerve, a decision support system designed to improve vehicle reservation operations at a university in the Philippines. The study addressed inefficiencies in vehicle reservations, including difficulties in assigning vehicles and drivers, limited resource utilization, and high transportation costs.

The researchers used a mixed-method approach that combined quantitative data analysis and qualitative insights. They developed three linear programming models and integrated them into a decision support system using Excel VBA. The system was designed to support vehicle and driver allocation and improve the processing of reservation requests.

The results showed that the optimized model achieved 100% reservation-processing efficiency and reduced the average reservation cycle time by 70.85%. It also improved vehicle utilization by 17.52%, with projected annual cost savings of ₱509,181.71. These findings demonstrate how automated decision support and optimization can improve vehicle reservation operations.

This study is relevant to the proposed **Calma Transportation Services: Implementation of a Cash Register and Reservation System** because both projects aim to improve vehicle reservation management and operational efficiency. However, Transerve focuses on optimizing vehicle and driver assignments, while the Calma system focuses on customer reservations, cash register transactions, payment records, and sales reports. The study provides a useful reference for improving reservation workflows and organizing vehicle-related operations. [30]

Abat, Buen, and Giron [31] developed an *Online Car Rental System* at Don Mariano Marcos Memorial State University–North La Union Campus in La Union, Philippines. The study aimed to make renting vehicles easier and more convenient for people who need transportation for travel, business trips, and other purposes.

The researchers developed the system and evaluated its acceptability among 50 respondents. They assessed the system based on functional suitability, performance efficiency, usability, reliability, security, and portability. They also used the t-test and analysis of variance (ANOVA) to examine differences in respondents' evaluations.

The results showed that the system was considered acceptable by the respondents. Security testing identified five medium-risk and eleven low-risk vulnerabilities. The researchers also found no significant differences in respondents' perceptions of the system's functional suitability, performance efficiency, usability, security, and reliability.

This study is relevant to the proposed **Calma Transportation Services: Implementation of a Cash Register and Reservation System** because both systems aim to improve car rental operations through digital technology. The study provides a reference for evaluating system quality, usability, and security. However, the Calma system also focuses on cash register transactions, payment records, and sales reports to support the business's financial management. [31]

Balintag et al. [32] developed SORMS, or Steffi's Online Reservation Management System, to address problems caused by the manual reservation process at Steffi's Garden Resort in the Philippines. The study aimed to make reservations more organized, reduce booking errors, and improve the efficiency of the resort's operations.

The researchers used the Agile Scrum methodology and involved 50 participants, including 18 IT professionals and 32 end-users, in developing and evaluating the system. SORMS provided a web-based platform with real-time booking and digital payment features, allowing reservations and payments to be handled more conveniently.

The results showed that the system received a score of 4.68 for software quality based on ISO 25010 and 4.83 under the Technology Acceptance Model (TAM). The researchers also reported 94% fewer booking errors and 80% faster processing compared with the manual process. However, compatibility was identified as an area that could be improved in future versions.

This study is relevant to the proposed **Calma Transportation Services: Implementation of a Cash Register and Reservation System** because both systems aim to improve reservation processing and reduce errors through digital technology. Although SORMS was developed for a resort, its real-time booking and digital payment features provide useful ideas for the Calma system. The proposed system extends this approach by focusing on vehicle reservations, cash register transactions, payment records, and sales reports. [32]

Montesines et al. [33] conducted a study titled *Development of an Online Reservation and Monitoring System for the Use of TWC Dormitory, Function Rooms and Vehicles* for the TESDA Women's Center in the Philippines. The study aimed to improve the organization's reservation processes and monitor the use of its facilities and vehicles more efficiently.

The researchers used an action research approach, gathering information through document reviews, interviews, focus group discussions, and direct observation. They worked with the system's users and administrators to identify requirements and develop a system that supported reservation management and monitoring.

The study identified the need to understand users' requirements, improve existing business processes, and involve process owners throughout system development. It also emphasized the importance of adequate resources, technical skills, project management, and continued system maintenance to support successful implementation.

This study is relevant to the proposed **Calma Transportation Services: Implementation of a Cash Register and Reservation System** because both systems involve vehicle reservations and aim to improve operational efficiency through digital technology. While the TESDA system focuses on reserving and monitoring

government facilities and vehicles, the Calma system focuses on commercial vehicle rentals, customer bookings, payment transactions, and sales records. The study provides a useful reference for designing reservation workflows and managing vehicle availability. [33]

Dalisay [34] developed a *Multi-platform Online Reservation System for Car Services* to address problems experienced by automotive service centers in managing customer appointments. The study identified issues in manual reservation processes, including duplicate bookings, long queues, reporting difficulties, and human errors. The proposed system aimed to make booking more accessible through web and mobile platforms.

The researcher used the modified Waterfall model to identify system requirements, design the application, and conduct testing. Interviews and observations were used to understand the existing reservation procedures and determine the system's necessary features. The application included separate modules for administrators, employees, and customers, with functions for user management, booking management, data management, and reports.

The testing results showed that the system was stable and received positive user feedback. Usability testing was conducted using the System Usability Scale for the web interface and the Questionnaire for User Interaction Satisfaction for the mobile interface. The results indicated that the system was usable, although some users suggested improvements to its features and mobile experience.

This study is relevant to the proposed **Calma Transportation Services: Implementation of a Cash Register and Reservation System** because both systems aim to improve reservation processing and reduce errors through digital technology. Although Dalisay [34] focused on automotive service appointments rather than vehicle rentals, the study provides useful ideas for booking management, user roles, reporting, and multi-platform access. The Calma system extends these concepts by including cash register transactions, payment records, and sales reports. [34]

Libunao et al. [35] developed TrialaTS, an online transportation booking system for residents of Barangay Triala, Guimba, Nueva Ecija, Philippines. The study aimed to address transportation problems such as long waiting times, unreliable services, and difficulty accessing available rides. The system was designed to make registration and booking more convenient while allowing barangay administrators to oversee transportation activities.

The researchers used a descriptive-developmental research design and the Agile Scrum methodology to develop and evaluate the system. They assessed its quality using ISO 25010 criteria, including functional suitability, usability, security, and reliability. They also used the Technology Acceptance Model (TAM) to evaluate users' perceived usefulness and ease of use.

The findings indicated that TrialaTS improved transportation accessibility, reduced waiting times, and made booking more convenient for residents. The study also reported positive user acceptance and benefits for drivers through a more structured booking process. The researchers recommended future improvements, including real-time tracking, greater scalability, and integration with other transportation networks.

This study is relevant to the proposed **Calma Transportation Services: Implementation of a Cash Register and Reservation System** because both systems aim to improve transportation bookings through digital technology. Although TrialaTS focuses on arranging local rides rather than renting vehicles, its online booking process and evaluation criteria provide useful guidance for the Calma system. The proposed system additionally focuses on vehicle availability, cash register transactions, payment records, and sales reports. [35]

Palmones and Maravilla [36] conducted a study titled “Evaluating Danao City's Transportation Sector: Towards a QR Code-Based Online Payment System.” The study examined transportation challenges in Danao City, Cebu, particularly fare inconsistencies, limited payment options, transaction security, and the lack of digital payment services.

The researchers used a descriptive research design and collected survey data from 96 respondents, including tourists, residents, and transportation providers. They examined users' experiences with local transportation services and their preferences for a proposed QR code-based online payment system.

The results showed dissatisfaction with fare inconsistencies, limited payment options, and transaction transparency. Security concerns involving theft and fraud received the highest weighted mean of 4.14. Additionally, 92.71% of respondents preferred features such as online booking and payment, highlighting the demand for more convenient digital transportation services.

This study is relevant to the proposed **Calma Transportation Services: Implementation of a Cash Register and Reservation System** because both projects address the need for more organized and convenient transportation transactions. While Palmones and Maravilla [36] focused on QR code-based payments for local transportation, the Calma system focuses on vehicle reservations, cash register transactions, payment records, and sales reports. The study supports the importance of convenient payment options and transparent transaction records in transportation services.[36]

The reviewed local studies highlight how digital technology can support transportation and rental operations in the Philippines. Albino and Acebedo [24] focused on a car rental management system with a scheduling algorithm, while Golo and Encarnacion [25] explored the development of an online vehicle rental management system for rental services in Siargao Island. De Jesus et al. [26] presented PARAM, a web-based booking and management system with a mobile tracking application for car rentals. Similarly, Marafo et al. [28] developed Arkila, a progressive web application for car rental management, while Abat et al. [31] examined an online car rental system. These studies demonstrate how digital platforms can support booking, scheduling, vehicle monitoring, and rental management.

Other local studies focus on reservation systems and digital solutions for different transportation-related needs. Montesines et al. [33] developed an online reservation and monitoring system for the use of vehicles and

other facilities at the TESDA Women's Center. Dalisay [34] focused on a multi-platform online reservation system for car services, while Libunao et al. [35] presented TrialaTS, a transportation system. Palmones and Maravilla [36] examined Danao City's transportation sector in relation to a QR code-based online payment system. These studies show how digital reservation and payment solutions can help organize service transactions and improve convenience for users.

Overall, the reviewed local studies provide a foundation for developing digital systems that support reservations, vehicle management, and transportation services. These studies are relevant to the proposed **Calma Transportation Services: Implementation of a Cash Register and Reservation System,** which aims to combine reservation management with cash register transactions, payment records, and sales reports. While the reviewed studies focus on areas such as booking, scheduling, vehicle tracking, and digital payments, the proposed system also emphasizes the recording of financial transactions and the organization of sales information. Integrating these functions into one platform aims to help Calma Transportation Services manage reservations and financial records in a more systematic and organized manner.

2.3 Related Literature

Web-Based Systems

Web-based systems have become one of the most significant technological innovations in modern information technology, providing organizations with efficient and accessible solutions for managing business operations and delivering services to users. A web-based system is a software application that operates through a web browser and can be accessed through a network or the internet without requiring installation on individual devices. These systems enable users to interact with applications and databases remotely, making them highly flexible, scalable, and convenient for both organizations and customers. As digital technology continues to advance, web-based systems have become widely adopted across various industries, including education, healthcare, banking, retail, tourism, transportation, and vehicle rental services.

The growing popularity of web-based systems can be attributed to their ability to provide real-time access to information and services. Unlike traditional desktop applications that are limited to specific computers or locations, web-based systems allow authorized users to access information from anywhere with an internet connection. This accessibility improves communication, coordination, and operational efficiency within organizations. Employees, administrators, and customers can perform transactions, retrieve information, and manage records through a centralized platform without being physically present in a particular location. As a result, web-based systems support more flexible and responsive business operations.

One of the primary advantages of web-based systems is their ability to automate business processes. Traditional manual systems often require employees to perform repetitive tasks such as recording transactions, updating records, calculating payments, and preparing reports. These processes consume significant amounts of time and increase the possibility of human error. Through automation, web-based systems can execute these tasks accurately and consistently while reducing administrative workload. Automated processing not only improves efficiency but also allows employees to focus on more important responsibilities that require critical thinking and customer interaction.

Another important feature of web-based systems is centralized data management. Information entered into the system is stored in a centralized database where it can be accessed, updated, and maintained by authorized users. Centralized databases help ensure data consistency and accuracy by eliminating duplicate records and reducing the risk of information loss. In service-oriented businesses, centralized data management enables efficient tracking of customer information, transactions, reservations, and operational records. This capability improves organizational productivity and supports better decision-making by providing accurate and updated information whenever needed.

Web-based systems also contribute significantly to improving customer satisfaction. Modern customers expect fast, convenient, and reliable services that can be accessed with minimal effort. Through web-based platforms, users can access services at any time and from any location, eliminating the need for unnecessary travel or lengthy waiting periods. Customers can view available services, make reservations, update information, and

receive confirmations instantly. These features enhance convenience and create a more positive user experience, which is essential for maintaining customer trust and loyalty.

In addition to improving accessibility and convenience, web-based systems offer enhanced security and data protection. Modern systems incorporate various security measures such as user authentication, password encryption, role-based access control, and database backup mechanisms. These features help protect sensitive information from unauthorized access, data breaches, and accidental loss. Organizations that implement secure web-based systems are better equipped to safeguard customer information and maintain the integrity of their business operations.

Scalability is another major benefit associated with web-based systems. As organizations grow and operational requirements increase, web-based applications can be expanded and upgraded without requiring significant changes to existing infrastructure. New features, additional users, and larger volumes of data can be accommodated more easily compared to traditional systems. This flexibility makes web-based systems highly suitable for businesses that anticipate future growth and changing operational demands.

Furthermore, web-based systems support efficient reporting and monitoring capabilities. Information stored within the database can be analyzed and transformed into meaningful reports that assist managers and administrators in evaluating business performance. Automated report generation allows organizations to monitor transactions, identify trends, assess operational efficiency, and make informed decisions. The availability of accurate and timely reports contributes to improved planning, resource allocation, and strategic development.

In transportation and vehicle rental services, web-based systems play a crucial role in improving reservation management, vehicle monitoring, customer record handling, and transaction processing. Through a centralized and automated platform, administrators can monitor vehicle availability in real time, process reservations efficiently, and maintain accurate customer information. Customers also benefit from faster and more convenient booking procedures, resulting in improved service quality and overall satisfaction.

The relevance of web-based systems to the proposed **Cash Register and Reservation System for Calma Transportation Services** is highly significant. The proposed system utilizes web-based technology to automate reservation handling, payment processing, customer record management, and report generation. By replacing manual processes with a computerized platform, the system aims to improve operational efficiency, reduce human errors, enhance data security, and provide better service to customers. Consequently, the implementation of a web-based system serves as a practical and effective solution for addressing the operational challenges currently experienced by Calma Transportation Services while supporting its long-term growth and development.

System Automation

System automation refers to the use of technology to perform tasks and processes with minimal human intervention. In modern organizations, automation has become a critical component in improving operational efficiency, reducing repetitive workloads, and enhancing service delivery. Through automated systems, businesses can execute routine activities such as data entry, record management, transaction processing, and report generation more accurately and consistently than manual methods. As technology continues to evolve, automation has transformed the way organizations operate by enabling faster, more reliable, and more efficient business processes.

The implementation of automation provides numerous benefits to service-oriented industries. Traditional manual operations often require employees to spend significant time performing repetitive tasks, which can increase the likelihood of human errors and reduce overall productivity. Automated systems eliminate many of these challenges by executing predefined procedures consistently and accurately. This allows employees to focus on more important responsibilities that require decision-making and customer interaction. Consequently, automation contributes to improved efficiency, better resource utilization, and enhanced organizational performance.

In transportation and vehicle rental businesses, automation plays an important role in managing reservations, customer records, vehicle availability, and payment transactions. Automated reservation systems can instantly update booking information, preventing scheduling conflicts and reducing administrative workload.

Similarly, automated payment processing ensures accurate calculations and faster transaction completion. These capabilities contribute to improved customer satisfaction and more efficient service delivery.

The proposed Cash Register and Reservation System incorporates automation to streamline reservation management, transaction processing, and record keeping. By reducing dependence on manual procedures, the system aims to improve operational efficiency, minimize errors, and provide a more reliable service experience for both administrators and customers.

Database Management

Database management is a fundamental component of modern information systems that involves the organization, storage, retrieval, and maintenance of data within a structured environment. As businesses generate increasing amounts of information, effective database management has become essential in ensuring that data remains accurate, accessible, and secure. Databases serve as centralized repositories where information can be stored systematically and accessed whenever needed by authorized users.

Traditional methods of storing information often rely on paper records, spreadsheets, and separate files. While these methods may be suitable for small-scale operations, they become inefficient as data volume increases. Manual record management can lead to data duplication, inconsistencies, loss of information, and difficulties in retrieving records. Database management systems address these challenges by providing structured storage mechanisms that improve data integrity and accessibility.

Centralized database systems allow organizations to maintain a single source of information that can be shared across multiple functions and departments. This ensures consistency and accuracy while reducing redundancy. In reservation-based businesses, databases play a crucial role in maintaining customer information, booking records, vehicle details, and payment histories. Real-time updates ensure that information remains current and available whenever needed.

The proposed system utilizes a centralized database to manage reservations, customer profiles, vehicles, and transactions efficiently. Through proper database management, the system ensures accurate information processing, improved accessibility, and enhanced operational performance.

User Interface Design

User Interface (UI) Design refers to the process of creating visual and interactive elements that enable users to communicate effectively with a system. A well-designed interface serves as the bridge between users and technology by presenting information in a clear, organized, and understandable manner. The success of any information system largely depends on how easily users can navigate and interact with its features.

Modern literature emphasizes that usability is one of the most important factors influencing user satisfaction and system acceptance. Systems that are difficult to understand or navigate often discourage users and reduce productivity. Effective user interface design focuses on simplicity, consistency, responsiveness, and accessibility. These principles help users complete tasks efficiently while minimizing confusion and errors.

In web-based systems, user interfaces must accommodate individuals with varying levels of technical expertise. Features such as intuitive navigation menus, clearly labeled functions, organized layouts, and responsive designs contribute to a positive user experience. A well-designed interface not only improves efficiency but also increases user confidence in using the system.

For the proposed Cash Register and Reservation System, user interface design is a critical consideration. The system aims to provide an accessible and user-friendly environment that enables administrators and staff members to manage reservations, payments, and records efficiently.

Cash Register Systems

Cash Register Systems are computerized solutions designed to record, calculate, and manage financial transactions within a business. These systems have evolved significantly from traditional mechanical cash registers into sophisticated digital platforms capable of handling sales, payments, inventory updates, and financial

reporting. In modern business environments, cash register systems play an essential role in ensuring transaction accuracy and financial accountability.

Manual transaction processing often involves handwritten receipts and manual calculations, which can lead to errors and inconsistencies. These mistakes may result in inaccurate financial records, customer disputes, and operational inefficiencies. Cash register systems address these issues by automating payment calculations and maintaining detailed transaction histories.

Modern cash register systems are commonly integrated with databases and reporting tools, allowing businesses to monitor sales activities and generate financial summaries efficiently. This capability supports better financial management and informed decision-making. Additionally, automated systems reduce administrative workload and improve transaction speed.

The proposed system integrates a cash register feature that automates payment calculations and transaction recording. This functionality ensures accurate financial processing while supporting efficient business operations.

System Development Life Cycle (SDLC)

The System Development Life Cycle (SDLC) is a structured methodology used in the planning, design, development, testing, implementation, and maintenance of information systems. SDLC provides a systematic approach that helps developers create reliable, efficient, and user-centered systems while minimizing development risks and ensuring project success.

The SDLC process typically begins with planning and requirements gathering, where developers identify organizational needs and system objectives. This is followed by analysis and design phases, during which system specifications and architectures are developed. Once the design is completed, developers proceed to implementation, testing, deployment, and maintenance activities.

One of the major advantages of SDLC is its ability to ensure that systems are developed according to user requirements and organizational goals. Structured development processes improve quality assurance and reduce the likelihood of system failures. Additionally, SDLC provides a framework for evaluating project progress and maintaining system performance after deployment.

The development of the proposed Cash Register and Reservation System follows SDLC principles to ensure that the final product meets functional requirements and supports efficient business operations.

Reservation Management System

Reservation Management Systems are computerized platforms designed to facilitate the scheduling, monitoring, and management of reservations in service-oriented businesses. These systems allow organizations to efficiently handle customer bookings while minimizing manual intervention. Traditionally, reservation processes relied on paper records, phone calls, and face-to-face transactions, which often resulted in scheduling conflicts, delayed confirmations, and data inaccuracies. With the advancement of information technology, reservation systems have evolved into web-based platforms capable of providing real-time updates and automated transaction processing.

Modern reservation systems provide numerous benefits to both service providers and customers. They improve operational efficiency by automating booking procedures, reducing paperwork, and maintaining accurate records of reservations. Customers benefit from faster access to services and the ability to make reservations at their convenience. In transportation-related industries, reservation systems play a significant role in ensuring that vehicle availability is properly monitored and that rental schedules are managed effectively.

Furthermore, reservation systems contribute to improved customer satisfaction by reducing waiting times and providing more reliable services. Through centralized databases, businesses can easily retrieve customer information, track reservation history, and generate reports that support decision-making. As a result, reservation management systems have become an essential component of modern service operations.

Management Information System

Management Information Systems (MIS) refer to organized systems that collect, process, store, and distribute information to support decision-making, coordination, and control within an organization. MIS serves as a bridge between technology and business operations by providing managers and employees with accurate and timely information necessary for efficient organizational performance.

In service-oriented businesses, MIS improves operational effectiveness by integrating various business functions into a centralized platform. These systems enable organizations to manage customer records, monitor transactions, generate reports, and analyze business performance more effectively. Through the use of computerized systems, businesses can reduce the time required to process information and improve the accuracy of operational activities.

The implementation of MIS also contributes to better communication and coordination among employees. Since information is stored in a centralized database, authorized users can easily access updated records whenever necessary. This reduces duplication of work, minimizes data inconsistencies, and ensures that business decisions are based on accurate information.

Furthermore, MIS supports strategic planning by providing managers with relevant reports and performance indicators. These reports help organizations identify trends, evaluate operational efficiency, and develop appropriate strategies for future growth. In transportation-related businesses, management information systems play a vital role in monitoring reservations, vehicle utilization, customer transactions, and financial records.

As businesses continue to embrace digital transformation, MIS remains an essential component for improving efficiency, enhancing productivity, and maintaining competitiveness in a rapidly changing business environment.

Customer Relationship Management

Customer Relationship Management (CRM) is a business strategy and technological approach that focuses on building and maintaining strong relationships between organizations and their customers. CRM

systems are designed to collect, organize, and analyze customer information to improve communication, customer service, and overall customer satisfaction.

In service industries, customer satisfaction is one of the most important factors influencing business success. Customers expect fast, reliable, and convenient services that meet their needs efficiently. CRM systems help organizations achieve these expectations by providing detailed information about customer preferences, transaction histories, and service interactions.

The use of CRM technology allows businesses to personalize their services and improve customer engagement. By maintaining accurate customer records, organizations can better understand customer needs and provide services that match their expectations. This leads to increased customer loyalty and stronger business relationships.

CRM systems also improve communication by enabling businesses to respond to customer inquiries, complaints, and feedback more effectively. Through organized customer databases, service providers can quickly access relevant information and resolve issues in a timely manner.

In the context of transportation and car rental services, CRM plays a significant role in managing customer reservations, monitoring rental history, and maintaining customer records. The integration of CRM principles into a reservation system can improve customer experience by providing more personalized, efficient, and reliable services.

Digital Transformation in Transportation Services

Digital transformation refers to the integration of digital technologies into various business processes to improve efficiency, productivity, and service quality. In recent years, transportation services have undergone significant digital transformation due to advances in information technology, internet connectivity, and mobile applications.

Traditional transportation businesses often relied on manual procedures for managing reservations, scheduling services, and recording transactions. These processes were frequently associated with delays, inefficiencies, and human errors. Through digital transformation, transportation companies have adopted computerized systems that automate many of these operational activities.

One of the major benefits of digital transformation is the ability to provide real-time information. Customers can easily access transportation services, check availability, make reservations, and receive updates through digital platforms. This improves convenience and enhances the overall customer experience.

For transportation service providers, digital technologies improve operational management by enabling real-time monitoring of resources, automated transaction processing, and centralized data management. These capabilities contribute to more efficient service delivery and better resource utilization.

Moreover, digital transformation supports business growth by allowing organizations to expand their services and reach a wider customer base. Web-based platforms and online reservation systems provide businesses with opportunities to improve accessibility while reducing operational costs.

As transportation services continue to evolve, digital transformation remains a critical factor in improving competitiveness, efficiency, and customer satisfaction within the industry.

Data Security and Privacy

Data security and privacy are important considerations in the development and implementation of information systems. As organizations increasingly rely on digital platforms to manage sensitive information, protecting data from unauthorized access, loss, and misuse has become a major priority.

Data security refers to the measures and technologies used to safeguard information against threats such as hacking, data breaches, malware attacks, and unauthorized modifications. Effective security practices ensure that information remains confidential, accurate, and accessible only to authorized users.

Privacy, on the other hand, focuses on protecting personal information and ensuring that customer data is collected, stored, and used responsibly. Organizations have a responsibility to maintain the confidentiality of customer information and comply with applicable data protection regulations.

In web-based systems, several security measures are commonly implemented to protect data. These include user authentication, password encryption, access control mechanisms, database security protocols, and regular system backups. Such measures reduce the risk of data loss and unauthorized access.

For transportation and car rental services, protecting customer information is particularly important because systems often store personal details, booking records, and payment information. Failure to secure this data can lead to financial losses, reputational damage, and reduced customer trust.

Therefore, incorporating data security and privacy measures into system design is essential for ensuring system reliability, protecting user information, and maintaining customer confidence. The proposed Cash Register and Reservation System recognizes the importance of these principles and aims to provide a secure environment for managing customer and transaction data.

2.4 Synthesis of the Reviewed Literature

The reviewed literature and studies, both foreign and local, collectively emphasize the growing importance and relevance of web-based systems in improving operational efficiency, accuracy, reliability, and overall service delivery across various industries, particularly in transportation and vehicle rental services. Across all examined studies, a consistent and recurring theme emerges: traditional manual processes are no longer sufficient to meet the increasing demands of modern service-oriented environments due to their inherent limitations in speed, accuracy, accessibility, and data management capabilities. As business operations become

more complex and customer expectations continue to rise, the need for automated and integrated digital solutions becomes increasingly essential.

Foreign studies strongly highlight that car rental systems and similar transportation platforms benefit significantly from the integration of automation, real-time communication, and advanced system functionalities such as GPS tracking, intelligent scheduling algorithms, mobile application integration, and dynamic booking systems. These technologies enhance operational workflows by reducing human intervention, minimizing errors, and ensuring that data is processed, updated, and synchronized in real time. Furthermore, foreign literature emphasizes that the integration of multiple system components—such as booking, payment processing, monitoring, and reporting—into a unified platform leads to improved decision-making, better resource management, and enhanced operational control. This ultimately results in higher levels of customer satisfaction and more efficient business operations.

On the other hand, local studies conducted in the Philippines reinforce and support these findings by demonstrating that many service-based businesses, including car rental and transportation services, still rely heavily on manual or semi-digital systems. These traditional approaches often result in operational inefficiencies such as booking conflicts, delayed transaction processing, inconsistent record keeping, and difficulty in tracking real-time vehicle availability. The reviewed local literature consistently shows that the implementation of web-based systems significantly improves transaction processing speed, data accuracy, and service accessibility. In addition, automation and centralized database systems are repeatedly identified as critical factors in reducing administrative workload, minimizing human error, and enhancing overall operational performance within local business environments.

Moreover, both foreign and local studies highlight the importance of system usability, accessibility, and structured development methodologies in ensuring system effectiveness and sustainability. Features such as user-friendly interfaces, responsive design, real-time updates, secure data handling, and integrated financial transaction systems are consistently recognized as essential components of an effective web-based application. The inclusion of cash register or payment management functionalities is also emphasized in related literature as a vital feature

for ensuring accurate financial tracking, transparent transaction processing, and improved accountability in business operations. Additionally, studies stress the importance of following structured development approaches such as Agile or SDLC to ensure systematic planning, continuous improvement, and successful system implementation.

Furthermore, system security and data integrity are also highlighted as important considerations in modern literature. As web-based systems handle sensitive information such as customer data, booking details, and financial transactions, implementing proper security measures such as authentication, authorization, and encrypted data storage is essential to protect system integrity and user privacy. Scalability and system performance are also important aspects, especially in systems that must accommodate increasing numbers of users and transactions over time without compromising speed or efficiency.

Overall, the synthesis of the reviewed literature and studies clearly indicates that the integration of automation, centralized database systems, real-time processing, secure data management, and financial transaction handling is essential in developing efficient, reliable, and user-centered service-based applications. These combined insights strongly support the development of the proposed Calma Transportation Services: Implementation of a Cash Register and Reservation System, as it aims to address the limitations of traditional manual processes by providing a unified, automated, secure, and user-friendly platform that enhances both operational efficiency and customer experience.

PHINMA UNIVERSITY OF ILOILO

College of Information Technology Education

CHAPTER 3

METHODOLOGY

3.1 Research Design

This study employed a **Developmental Research Design** as the primary research approach in developing the proposed *Cash Register and Reservation System for Calma Transportation Services*. This type of research design is commonly used in information technology and software development projects because it focuses on creating, improving, and evaluating a product, system, or application that addresses existing operational problems. Unlike other research designs that mainly concentrate on describing or analyzing a situation, developmental research emphasizes the development of practical solutions that can be implemented in real-world environments.

The researchers selected this research design because the main objective of the study is not only to examine the current reservation and transaction management practices of Calma Transportation Services but also to develop a computerized system that can improve these processes. At present, many businesses still encounter challenges related to manual record keeping, reservation monitoring, customer management, and transaction processing. These challenges often result in delays, inaccuracies, data redundancy, and difficulties in generating reports. Through developmental research, the researchers were able to systematically identify these concerns and develop a technology-based solution that addresses the specific needs of the business.

The developmental research process began with a thorough assessment of the existing operational procedures used by Calma Transportation Services. The researchers conducted interviews and observations to gain a clear understanding of how reservations are processed, how customer information is recorded, how transactions are handled, and how business records are maintained. This initial investigation enabled the researchers to identify weaknesses and inefficiencies within the current workflow and determine areas that required improvement through automation.

After gathering and analyzing the necessary information, the researchers proceeded with the design and development of the proposed system. The development process involved identifying system requirements, designing system components, creating the database structure, developing user interfaces, implementing system functionalities, and conducting testing procedures. Each stage was carefully planned and executed to ensure that

the final product would effectively support the operational requirements of the business and provide a more efficient method of managing reservations and transactions.

Another important characteristic of developmental research is its emphasis on continuous improvement throughout the development process. As the system was being developed, feedback and observations were continuously gathered to evaluate whether the system met its intended objectives. Necessary modifications and enhancements were implemented whenever issues or limitations were identified. This iterative process allowed the researchers to improve the overall quality, functionality, and usability of the system before its final evaluation.

Furthermore, developmental research encourages the integration of theoretical knowledge and practical application. In this study, concepts and principles related to information systems, database management, software engineering, and web application development were applied to create a system that addresses actual business challenges. By combining academic knowledge with practical implementation, the researchers were able to develop a solution that is both technically sound and operationally effective.

The use of developmental research design also supports the evaluation of the completed system. After development, the proposed Cash Register and Reservation System underwent testing and assessment to determine its effectiveness, reliability, usability, and functionality. The results of these evaluations provided valuable insights into the strengths of the system and identified potential areas for future improvement. This ensured that the developed application met the expectations of its intended users and fulfilled the objectives established at the beginning of the study.

Through the application of Developmental Research Design, the researchers were able to follow a structured and systematic process from problem identification to solution development and evaluation. This approach contributed significantly to the successful creation of a web-based Cash Register and Reservation System capable of improving reservation management, transaction processing, record keeping, and overall operational efficiency within Calma Transportation Services. Ultimately, the study demonstrates how technology can be utilized as a practical tool for enhancing business operations and delivering better service to customers.

3.2 System Development Methodology

System Development Methodology

The researchers utilized the **Agile Software Development Methodology** as the primary methodology for the development of the proposed *Cash Register and Reservation System for Calma Transportation Services*. Agile is a widely recognized approach in software development that provides a flexible and iterative process for creating information systems. It consists of several stages that guide developers through requirements gathering, system design, development, testing, deployment, and continuous improvement. By following this methodology, the researchers were able to develop the system through repeated cycles of development, testing, evaluation, and improvement, ensuring that the system continuously addressed the operational needs of Calma Transportation Services.

The selection of Agile was based on its ability to provide flexibility throughout the development process. Since the proposed system involves managing reservations, customer records, vehicle availability, and financial transactions, a flexible methodology was necessary to accommodate changes and improvements based on user feedback. Agile also allows developers to identify problems early in the development process and make necessary adjustments before proceeding to the next iteration. Through the use of this methodology, the researchers were able to develop a system that is reliable, efficient, and capable of meeting the operational needs of Calma Transportation Services.

Requirements Analysis Phase

The requirements analysis phase served as the foundation of the development process. During this stage, the researchers conducted preliminary investigations to gain a thorough understanding of the current operational procedures of Calma Transportation Services. The objective was to identify existing challenges and determine how technology could be utilized to improve business operations. Interviews and observations were conducted to

gather information regarding reservation handling, customer management, vehicle scheduling, transaction processing, and record keeping.

The researchers carefully examined the limitations of the current system and identified several areas that required improvement. Common issues included delays in processing reservations, difficulties in monitoring vehicle availability, manual recording of transactions, and challenges in maintaining accurate business records. These findings helped establish the objectives of the study and provided direction for the development of the proposed system.

In addition, functional and non-functional requirements were identified based on the needs of the business and its intended users. Proper requirements analysis allowed the researchers to establish realistic goals and ensure that the system would remain focused on addressing the identified operational problems.

System Design Phase

Once the requirements were identified, the researchers proceeded to the system design phase. This stage focused on creating a detailed blueprint that would guide the development of the system. The design process involved determining how the system would operate, how information would be stored, and how users would interact with the application.

Several design tools and techniques were utilized during this phase, including Data Flow Diagrams (DFD), Entity Relationship Diagrams (ERD), Use Case Diagrams, database schemas, and interface prototypes. These diagrams provided visual representations of the system's structure and functionality, making it easier to understand how different components would interact with one another.

The researchers also focused on designing an intuitive and user-friendly interface. Since both administrators and customers would interact with the system, special consideration was given to ensuring that menus, forms, buttons, and navigation features were easy to understand and use. The system design was also reviewed and improved based on feedback to ensure that it properly reflected the identified requirements.

Development Phase

The development phase involved transforming the approved system design into a functional web-based application. During this stage, the researchers utilized various programming languages, frameworks, and development tools to implement the planned features and functionalities of the system.

The frontend interface was developed to provide users with an interactive environment where they could make reservations, view booking information, manage accounts, and access available services. At the same time, the backend system was developed to handle business logic, data processing, authentication, and communication with the database.

Several core features were implemented during this phase, including reservation management, customer account management, vehicle monitoring, payment recording, report generation, and administrative controls. The researchers developed the system in manageable parts and continuously checked each feature to determine whether it functioned according to the established requirements. Changes and improvements were made whenever issues or new requirements were identified.

This phase represented an important part of the project because it required translating the system designs and requirements into actual software components. Through the iterative development process, the researchers were able to gradually build and improve the system until the required functionalities were achieved.

Testing Phase

After each major feature was developed, testing procedures were conducted to ensure that all functionalities worked as intended. The primary purpose of testing was to identify errors, defects, and inconsistencies that could affect the performance of the system.

The researchers performed different testing activities to verify the accuracy of system outputs and ensure that all modules functioned properly. Reservation processing, transaction recording, report generation, user authentication,

and database operations were carefully examined. The testing process also included evaluating how the system responded to various user inputs and operational scenarios.

Whenever errors or issues were discovered, corrective actions were immediately taken. The affected components were revised and tested again until the desired results were achieved. This iterative testing process allowed the researchers to continuously improve the system's stability, reliability, and functionality.

Deployment Phase

Following successful testing of the developed features, the system proceeded to the deployment phase. During this stage, the completed application was prepared and introduced to its intended users for demonstration and evaluation purposes. Administrators and staff members were given the opportunity to interact with the system and perform actual tasks using the developed application.

The deployment phase allowed users to experience the features of the system in a realistic environment. Users evaluated the system's performance, functionality, ease of use, and overall effectiveness in supporting business operations. Their feedback provided valuable information that could be used to identify areas requiring further improvement.

This phase was important because it allowed the researchers to determine whether the developed system could successfully meet user expectations and address the operational problems identified during the earlier stages of the project.

System Review and Feedback (Continuous Activity)

The system review and feedback phase focused on evaluating the developed system based on the experiences and comments of its intended users. The researchers reviewed the system's functionality, usability, performance, and effectiveness in supporting the operations of Calma Transportation Services.

Feedback from users was analyzed to determine whether additional changes or improvements were necessary. Issues identified during the review were documented and addressed by the researchers whenever applicable. The

system was then tested again after modifications to ensure that the changes did not negatively affect other functionalities.

This phase reflects the iterative nature of Agile methodology because development did not end after the initial implementation. Instead, the researchers continuously reviewed the system and considered feedback to improve its overall quality and suitability for the intended users.

Iteration and Improvement (Continuous Activity)

The iteration and improvement phase involved making necessary changes to the system based on the results of testing, user feedback, and system review. The researchers revisited certain parts of the system whenever improvements were required and repeated the appropriate development and testing activities.

Improvement activities included correcting software errors, enhancing existing features, improving the user interface, and adjusting system functions based on identified requirements. Through repeated iterations, the researchers were able to refine the system and ensure that its features remained aligned with the operational needs of Calma Transportation Services.

Regular iteration and improvement helped ensure that the system remained reliable, efficient, and user-friendly. It also provided the researchers with an opportunity to continuously enhance the application throughout the development process.

Through the use of the Agile Software Development Methodology, the researchers were able to follow a flexible and iterative process in developing the proposed Cash Register and Reservation System. These activities supported the Agile development process, while continuous testing, feedback, and improvement helped produce a web-based solution designed to improve efficiency, accuracy, and overall service quality within Calma Transportation Services.

3.3 System Architecture

The proposed Cash Register and Reservation System for Calma Transportation Services utilized a client-server architecture, a widely used system structure in modern web applications. This architecture was selected because it provides an efficient way of managing communication between users, the application itself, and the database where important records are stored. Through this approach, data can be processed accurately, stored securely, and retrieved efficiently whenever needed. The architecture also supports multiple users accessing the system simultaneously without significantly affecting overall performance.

The client-server architecture consists of three major components: the **client side**, the **server side**, and the **database layer**. Each component performs a specific function that contributes to the overall operation of the system. By separating responsibilities among these components, the system becomes easier to manage, maintain, and improve in the future.

Client Side

The client side serves as the part of the system that users directly interact with. It provides the graphical user interface through which customers, administrators, and staff members can access the system's various features and services. Since the proposed system is web-based, users can access it using devices such as desktop computers, laptops, tablets, or smartphones with an internet connection and a compatible web browser.

For customers, the client side allows them to browse available transportation services and vehicles, submit reservation requests, view reservation details, and monitor the status of their bookings. The interface is designed to be simple, organized, and easy to navigate, ensuring that users can perform tasks efficiently even with minimal technical knowledge.

For administrators and staff members, the client interface provides access to management functions such as monitoring reservations, updating vehicle information, recording transactions, managing customer accounts,

and generating reports. Through these features, authorized personnel can efficiently oversee daily operations and ensure that business processes run smoothly.

To improve accessibility and user experience, the interface is designed to be responsive. This means that the layout and functionality of the system automatically adjust according to the screen size and device being used. As a result, users can conveniently access the system from various devices without experiencing usability issues.

Server Side

The server side functions as the central processing unit of the system. It is responsible for handling requests received from users, processing information according to established business rules, and communicating with the database. Whenever a user performs an action, such as creating a reservation or recording a payment, the request is sent to the server for processing.

The server validates all incoming data to ensure accuracy and consistency before any information is stored in the database. For example, when a customer submits a reservation request, the server checks vehicle availability, verifies booking schedules, calculates applicable charges, and confirms whether the reservation can proceed. This process helps prevent scheduling conflicts and ensures that only valid transactions are recorded.

In addition, the server manages user authentication and access control. It verifies login credentials and determines the level of access granted to each user. This security mechanism ensures that sensitive functions, such as modifying reservations or generating financial reports, can only be performed by authorized personnel.

The server also handles communication between the user interface and the database. By acting as an intermediary, it helps protect sensitive information and reduces the risk of unauthorized access or data manipulation.

Database Layer

The database layer serves as the storage component of the system. It contains all information required for the daily operations of Calma Transportation Services. This includes customer records, reservation details, vehicle information, transaction histories, payment records, user accounts, and system-generated reports.

The database is designed using a relational structure to ensure that data is organized logically and efficiently. Related information is stored in separate tables and connected through relationships, reducing duplication and improving data consistency. This structure allows the system to retrieve information quickly while maintaining accuracy and integrity.

Whenever users create reservations, update records, or perform transactions, the information is automatically stored in the database. Similarly, when reports or booking details are requested, the database provides the required information to the server for processing and display.

The database also plays an important role in maintaining business records. Since information is stored electronically, administrators can easily search, retrieve, and generate reports without manually reviewing paper-based documents. This significantly improves productivity and reduces the likelihood of errors associated with manual record keeping.

System Communication Process

The interaction between the client side, server side, and database follows a continuous communication cycle. When a user submits a request through the client interface, the request is transmitted to the server. The server then processes the request and communicates with the database whenever information needs to be stored, modified, or retrieved.

After the necessary operations have been completed, the server sends the processed results back to the client side, where the information is displayed to the user. This process occurs within seconds, allowing users to receive immediate feedback regarding their actions.

For example, when a customer creates a reservation, the system follows the following sequence:

1. The customer enters reservation details through the web interface.
2. The information is sent to the server for validation.
3. The server checks vehicle availability and booking schedules.
4. The server records the reservation in the database.
5. The database confirms successful storage of the information.
6. The server sends a confirmation message to the user.
7. The reservation details are displayed on the customer's screen.

This workflow ensures that all transactions are processed accurately and efficiently while maintaining data consistency throughout the system.

Security and Scalability

One of the major advantages of using a client-server architecture is enhanced security. Sensitive operations such as user authentication, reservation validation, and transaction processing are performed on the server side rather than on the user's device. This reduces the possibility of unauthorized modifications and helps protect important business information.

The architecture also supports scalability, allowing the system to accommodate future growth. As the business expands and the number of users increases, additional resources can be added to the server and database without requiring major changes to the overall structure of the application. This flexibility ensures that the system remains capable of supporting future operational demands.

Overall, the client-server architecture provides a strong and reliable foundation for the proposed Cash Register and Reservation System. By separating the responsibilities of the client interface, server processing, and database management, the system can operate efficiently while maintaining security, reliability, and data accuracy. This architectural design supports the primary objective of improving reservation management, transaction processing, and record keeping within Calma Transportation Services while providing a convenient and user-friendly experience for both customers and administrators.

3.4 Tools and Technologies Used

The successful development of the proposed *Cash Register and Reservation System for Calma Transportation Services* requires the use of various software tools, programming technologies, and development environments. These technologies were carefully selected based on their capability to support the requirements of the system, including functionality, performance, security, scalability, and ease of maintenance. By utilizing modern web development technologies, the researchers were able to create a system that is efficient, user-friendly, and capable of supporting the operational needs of the business.

The development process involves both frontend and backend technologies, along with database management systems and development tools. Each technology plays a specific role in ensuring that the system operates effectively and delivers a smooth user experience.

Frontend

The frontend refers to the part of the system that users directly interact with through their web browsers. It is responsible for presenting information, collecting user inputs, and providing an interactive and visually appealing experience.

React with TypeScript

The researchers utilized **React** as the primary JavaScript library for developing the user interface of the system. React is widely recognized for its ability to create dynamic and responsive web applications through the use of reusable components. This approach allows developers to organize the application into smaller and manageable sections, making development more efficient and improving maintainability.

React was selected because it enables faster rendering of web pages and provides a seamless user experience. Since the proposed system involves frequent interactions such as viewing reservations, managing vehicle information, and processing transactions, React helps ensure that updates are displayed efficiently without requiring the entire page to reload.

Alongside React, **TypeScript** was used to improve code quality and reliability. TypeScript is an extension of JavaScript that introduces static typing, which helps developers identify potential errors during development rather than during system execution. This feature contributes to the stability of the application by reducing programming mistakes and improving overall code organization.

The combination of React and TypeScript provides a robust development environment that supports the creation of a modern, efficient, and scalable web application.

Vite

To support frontend development, the researchers utilized **Vite** as the build and development tool for the project. Vite is designed to provide faster startup times and quicker updates during development compared to traditional web development tools.

One of the primary advantages of Vite is its ability to improve developer productivity. It allows developers to immediately view changes made to the application without lengthy compilation processes. This feature significantly reduces development time and enables the researchers to test and refine system functionalities more efficiently.

Additionally, Vite produces optimized production builds, helping improve the overall performance and loading speed of the final application. Faster loading times contribute to a better user experience, especially for customers accessing the reservation system through different devices and internet connections.

Bootstrap

The researchers also utilized **Bootstrap** as the primary front-end framework for designing the visual appearance of the system. Bootstrap provides a collection of pre-designed user interface components such as buttons, forms, navigation menus, tables, and responsive layouts.

Bootstrap was selected because it simplifies the design process while maintaining consistency throughout the application. By using Bootstrap components, the researchers were able to create a professional-looking interface without developing every design element from scratch.

Another important advantage of Bootstrap is its responsive design capability. This ensures that the system automatically adjusts to different screen sizes, allowing users to access the application comfortably from desktop computers, laptops, tablets, and smartphones. Responsive design improves accessibility and ensures a positive user experience regardless of the device being used.

Backend

The backend serves as the operational core of the system. It processes user requests, executes business logic, validates information, and communicates with the database.

Node.js

For server-side development, the researchers utilized **Node.js** as the backend runtime environment. Node.js enables JavaScript code to run on the server, making it possible to handle system operations efficiently and manage communication between users and the database.

Node.js was selected because of its ability to process multiple requests simultaneously without significantly affecting system performance. This capability is particularly important for reservation systems where multiple users may attempt to access services at the same time.

Within the proposed system, Node.js is responsible for handling reservation requests, processing transactions, managing user authentication, validating information, and generating reports. It serves as the central component that coordinates interactions between the frontend and the database.

Furthermore, Node.js is known for its scalability and efficiency. As the number of users grows, the system can continue to perform effectively without major modifications to its architecture. This makes it a suitable technology for both current operations and future expansion.

Database

The database serves as the central repository for all information used within the system. Proper database management is essential for maintaining data accuracy, consistency, and security.

MySQL

The proposed system utilizes **MySQL** as its relational database management system. MySQL is one of the most widely used database platforms due to its reliability, efficiency, and ability to manage large amounts of structured data.

The database stores various types of information required for system operations, including customer records, reservation details, vehicle information, payment transactions, user accounts, and generated reports. By organizing data into related tables, MySQL helps reduce redundancy and ensures consistency across the entire system.

One of the major advantages of MySQL is its ability to retrieve information quickly and accurately. This capability is particularly important when users need to access reservation records, verify transaction histories, or generate business reports.

MySQL also provides security features that help protect sensitive information from unauthorized access. These features contribute to maintaining data integrity and ensuring that business records remain accurate and secure.

Development Environment

To facilitate efficient system development, the researchers utilized modern software development tools that support coding, debugging, testing, and project management activities.

Visual Studio Code (VS Code)

The primary Integrated Development Environment (IDE) used for the project was **Visual Studio Code (VS Code)**. VS Code is a lightweight yet powerful code editor widely used by professional developers and students alike.

VS Code provides numerous features that improve development efficiency, including syntax highlighting, code completion, debugging tools, extension support, and integrated version control. These features help developers write cleaner code, identify errors quickly, and maintain organized project files.

The software also supports various programming languages and frameworks used in the project, making it an ideal development environment for full-stack web application development. Through VS Code, the researchers were able to manage both frontend and backend components of the system within a single workspace.

3.5 Data Gathering Techniques

The success of the proposed *Cash Register and Reservation System for Calma Transportation Services* largely depends on the accuracy and relevance of the information gathered during the initial stages of the study. To ensure that the system addresses the actual needs and challenges faced by the business, the researchers employed appropriate data gathering techniques that provided valuable insights into the existing operational procedures. These techniques enabled the researchers to obtain reliable information regarding reservation management, customer transactions, vehicle monitoring, and record-keeping practices.

Data gathering is an important part of system development because it allows researchers to understand the current situation before proposing improvements. By collecting information directly from individuals involved in daily business operations, the researchers were able to identify existing problems, determine user requirements, and develop solutions that are practical and applicable to the organization.

For this study, the researchers utilized two primary data gathering techniques: **Interviews** and **Observation**. These methods were selected because they provide both qualitative and firsthand information regarding the actual processes and challenges experienced by Calma Transportation Services.

Interviews

Interviews were conducted to gather detailed information from individuals directly involved in the operations of Calma Transportation Services. This method allowed the researchers to communicate with business owners, employees, and other stakeholders who possess valuable knowledge regarding the organization's reservation procedures, transaction management, and customer service activities.

The interview process involved asking a series of prepared questions related to the current methods used in handling reservations, recording payments, managing customer information, and monitoring vehicle availability. In addition to structured questions, follow-up questions were also asked whenever clarification or additional details were needed. This approach enabled the researchers to gain a deeper understanding of the operational challenges encountered by the business.

Through the interviews, several important concerns were identified. These included difficulties in managing reservation records, delays in confirming bookings, challenges in monitoring vehicle availability, and the possibility of errors resulting from manual data entry. Stakeholders also expressed the need for a more organized and efficient system that could simplify daily operations and reduce the workload associated with manual processes.

The information gathered from interviews played a significant role in determining the features and functionalities that would be included in the proposed system. By listening to the experiences and suggestions of actual users, the researchers were able to develop a solution that directly addresses the needs of the organization.

Furthermore, interviews provided insights into user expectations regarding system usability and performance. Understanding these expectations helped ensure that the proposed application would not only function correctly but would also be easy to use and beneficial to its intended users. The interview process therefore served as a valuable source of information that guided both the design and development phases of the project.

Observation

In addition to interviews, the researchers utilized observation as a data gathering technique to obtain firsthand information regarding the actual workflow and business operations of Calma Transportation Services. Observation involves directly examining how tasks and activities are performed within the organization without relying solely on verbal descriptions from participants.

During the observation process, the researchers carefully monitored how reservations were received, recorded, and processed. They also observed how customer information was managed, how transactions were documented, and how vehicle availability was tracked. By witnessing these activities firsthand, the researchers gained a more accurate understanding of the operational environment and identified issues that may not have been fully revealed during interviews.

Observation allowed the researchers to identify inefficiencies within the existing process. For example, manual recording procedures often required additional time and effort, increasing the possibility of human errors and inconsistencies in data management. The researchers also observed situations where retrieving information from records became time-consuming due to the absence of a centralized system.

Another advantage of observation was the ability to verify information gathered during interviews. By comparing stakeholder responses with actual workplace practices, the researchers were able to ensure the

accuracy and reliability of the collected data. This helped strengthen the overall validity of the study and provided a more comprehensive understanding of the organization's needs.

Additionally, observation enabled the researchers to identify opportunities for automation and process improvement. By understanding how employees performed their daily tasks, the researchers were able to determine which activities could benefit most from computerization. These observations contributed significantly to the design of system features intended to improve efficiency, reduce errors, and streamline business operations.

Importance of Combining Interviews and Observation

The researchers recognized that relying on a single data gathering technique may not provide a complete picture of the organization's operational requirements. For this reason, interviews and observation were used together to obtain more comprehensive and reliable information.

Interviews provided detailed explanations, opinions, and suggestions from stakeholders, while observation allowed the researchers to witness actual workplace practices and verify the accuracy of the information collected. The combination of these methods enabled the researchers to examine the problem from multiple perspectives and develop a more accurate understanding of the organization's needs.

Furthermore, using multiple data gathering techniques increased the credibility of the study. Information obtained through interviews could be validated through observation, reducing the possibility of misunderstandings or incomplete data. This approach ensured that the proposed system was developed based on factual information rather than assumptions.

The integration of these techniques also helped the researchers identify both technical and operational requirements for the system. As a result, the proposed Cash Register and Reservation System was designed to address real-world challenges while supporting the daily activities of Calma Transportation Services.

Overall, the use of interviews and observation provided the researchers with sufficient information to analyze existing processes, identify operational issues, and determine the most appropriate features for the

proposed system. These data gathering techniques served as an essential foundation for the successful development of a web-based solution aimed at improving reservation management, transaction processing, record keeping, and overall service efficiency within Calma Transportation Services.

3.6 Testing Procedures

Testing is a crucial phase in system development because it ensures that the proposed system functions according to its intended purpose and meets the requirements identified during the earlier stages of development. It allows researchers to identify errors, inconsistencies, and technical issues before the system is fully implemented. Through proper testing, the reliability, functionality, and performance of the system can be evaluated and improved to provide a better experience for its intended users.

For the proposed *Cash Register and Reservation System for Calma Transportation Services*, testing was conducted to verify that all system components worked correctly and that the application was capable of supporting the operational needs of the business. The testing process focused on evaluating the system's ability to manage reservations, process transactions, maintain records, and provide accurate information to users.

The researchers conducted two major testing procedures, namely **System Testing** and **User Acceptance Testing (UAT)**. These testing methods were selected because they provide comprehensive evaluation of both the technical performance of the system and the satisfaction of its intended users.

System Testing

System Testing was conducted to evaluate the overall functionality of the developed application as a complete and integrated system. This testing procedure focused on determining whether all modules and

components worked together properly and whether the system could perform its intended functions without errors or interruptions.

During this phase, the researchers tested various features and functionalities of the system, including user registration, login authentication, reservation management, vehicle information management, payment recording, report generation, and account administration. Each feature was carefully examined to ensure that it produced the expected results and handled user inputs correctly.

The researchers also tested how information moved between the frontend interface, backend processes, and database. This was done to verify that data was accurately transmitted, stored, retrieved, and displayed whenever users performed actions within the system. Proper communication among system components is essential to maintaining data consistency and ensuring reliable operation.

Several test scenarios were performed to evaluate the system under different conditions. These scenarios included valid and invalid user inputs, multiple reservation requests, transaction processing activities, and report generation procedures. By conducting these tests, the researchers were able to identify potential issues such as incorrect calculations, duplicate records, data retrieval errors, and interface inconsistencies.

Whenever problems were detected, corrective actions were implemented immediately. The affected components were modified and retested until they functioned correctly. This process helped ensure that the final version of the system was stable, accurate, and capable of supporting daily business operations effectively.

Another important objective of system testing was to evaluate the overall performance of the application. The researchers examined whether the system responded quickly to user requests and whether it could handle multiple operations without experiencing significant delays. Performance testing contributed to ensuring a smooth and efficient user experience.

Through comprehensive system testing, the researchers gained confidence that the proposed application met the required specifications and was ready for evaluation by actual users.

User Acceptance Testing (UAT)

Following the completion of system testing, the researchers conducted **User Acceptance Testing (UAT)** to determine whether the developed application satisfied the needs and expectations of its intended users. User Acceptance Testing is considered one of the most important stages of system evaluation because it involves actual users who will eventually utilize the system in real-world situations.

For this study, administrators, staff members, and selected individuals familiar with the operations of Calma Transportation Services participated in the evaluation process. Participants were given access to the system and were asked to perform various tasks such as creating reservations, managing bookings, recording transactions, viewing reports, and navigating different sections of the application.

The purpose of this testing phase was to assess whether users could successfully perform their required tasks using the system. It also allowed the researchers to evaluate the usability, functionality, and overall effectiveness of the application from the perspective of its end users.

During User Acceptance Testing, participants were encouraged to provide feedback regarding their experience with the system. They were asked to identify any difficulties encountered while using the application and to provide suggestions for improvement. Feedback was collected through evaluation forms, interviews, and informal discussions.

Particular attention was given to factors such as ease of use, interface design, accessibility of features, system responsiveness, and overall user satisfaction. The researchers analyzed the feedback received and used it to determine whether additional modifications or enhancements were necessary before finalizing the system.

User Acceptance Testing also helped validate whether the objectives established at the beginning of the study had been successfully achieved. If users were able to complete tasks efficiently and expressed satisfaction with the system's performance, it indicated that the application effectively addressed the operational challenges identified during the data gathering stage.

Importance of Testing in System Development

Testing plays a significant role in ensuring the quality and success of any software application. Without proper testing, errors and system failures may remain undetected, potentially affecting business operations and user satisfaction. By conducting thorough testing procedures, the researchers were able to identify weaknesses, improve system performance, and ensure that all functionalities operated as intended.

The testing process also contributes to the reliability and credibility of the study. Demonstrating that the system has undergone systematic evaluation provides assurance that it can perform effectively in actual operational environments. Furthermore, testing allows researchers to gather valuable feedback that can be used to improve future versions of the application.

For the proposed Cash Register and Reservation System, testing served as an essential step in validating the effectiveness of the developed solution. It ensured that the application was capable of improving reservation management, transaction processing, record keeping, and overall operational efficiency within Calma Transportation Services.

3.7 Evaluation Criteria

After the development and testing of the proposed Cash Register and Reservation System for Calma Transportation Services, the system underwent evaluation to determine its effectiveness, quality, and suitability for actual business operations. The evaluation process is essential because it allows the researchers to measure whether the developed application successfully meets the objectives of the study and addresses the operational challenges identified during the data gathering phase.

The evaluation focused on key software quality attributes that are commonly used in assessing information systems. These criteria helped determine how well the system performs its intended functions, how

easily users can interact with it, and how dependable it is during daily operations. Through these evaluation measures, the researchers could identify the strengths of the system as well as areas that may require further improvement.

The proposed system was evaluated based on the following criteria: Functionality, Usability, and Reliability.

Functionality

Functionality refers to the ability of the system to perform its intended tasks accurately, efficiently, and completely. This criterion evaluates whether all required features and functions are working according to the specifications established during the planning and analysis stages of the project.

For the proposed Cash Register and Reservation System, functionality focuses on the system's ability to manage reservations, process transactions, maintain customer records, monitor vehicle availability, and generate reports. Each feature must operate correctly and provide accurate outputs based on user inputs.

The evaluation of functionality also determines whether the system effectively supports the daily workflow of Calma Transportation Services. For example, when a reservation is submitted, the system should accurately record the booking details, verify availability, and update the database without errors. Similarly, transaction records must be processed and stored correctly to ensure accurate financial reporting.

In addition, functionality evaluates the completeness of the system. This means that all required features identified during the requirement-gathering phase must be present and fully operational. Missing or malfunctioning features may negatively affect the efficiency and effectiveness of business operations.

A highly functional system enables users to perform their tasks more efficiently, minimizes operational errors, and contributes to improved service quality. Therefore, functionality serves as one of the most important indicators of the overall success of the proposed application.

Usability

Usability refers to the ease with which users can learn, understand, and operate the system. This criterion focuses on the overall user experience and evaluates whether the application is designed in a way that allows users to accomplish tasks efficiently and comfortably.

The proposed system is intended to be used by administrators, staff members, and customers with varying levels of technical knowledge. Because of this, the interface must be simple, organized, and user-friendly. Users should be able to navigate through different sections of the system without confusion or difficulty.

Several aspects are considered when evaluating usability. These include the clarity of menus and navigation options, the accessibility of features, the responsiveness of the interface, and the overall appearance of the application. Forms, buttons, and system messages should be easy to understand and guide users throughout the reservation and transaction processes.

Usability also measures how quickly users can become familiar with the system. A well-designed application should require minimal training and allow users to complete tasks efficiently even during their first interaction with the software. Reducing complexity and improving accessibility can significantly increase user satisfaction and productivity.

Furthermore, usability plays an important role in the acceptance of the system. Even if a system contains numerous features, users may hesitate to use it if it is difficult to understand or navigate. Therefore, ensuring a positive user experience is essential for the successful implementation of the proposed application.

Through usability evaluation, the researchers can determine whether the system provides a convenient and efficient platform for managing reservations, transactions, and business records.

Reliability

Reliability refers to the ability of the system to operate consistently and accurately over an extended period of time without experiencing failures, errors, or interruptions. This criterion evaluates the stability and dependability of the application during actual use.

For the proposed Cash Register and Reservation System, reliability is particularly important because business operations depend heavily on accurate records and uninterrupted access to information. Any system failure or data inconsistency could potentially affect reservation schedules, financial records, and customer satisfaction.

The evaluation of reliability focuses on the system's ability to maintain consistent performance under different operating conditions. The system should continue functioning properly even when handling multiple transactions, processing numerous reservations, or serving multiple users simultaneously.

Reliability also examines data accuracy and integrity. Information stored within the database must remain complete, consistent, and free from corruption. Reservation details, customer information, payment records, and generated reports should always reflect accurate and updated information.

Another important aspect of reliability is error handling. The system should be capable of identifying invalid inputs and responding appropriately without causing application crashes or loss of data. Effective error management contributes to a more stable and dependable user experience.

A reliable system increases user confidence and ensures that business operations can continue smoothly without unnecessary disruptions. For this reason, reliability serves as a critical factor in evaluating the overall quality of the developed application.

System Evaluation

System evaluation is an essential component of software development because it provides measurable evidence regarding the effectiveness of the proposed solution. Through evaluation, the researchers can determine whether the developed system successfully meets user requirements and fulfills the objectives established at the beginning of the study.

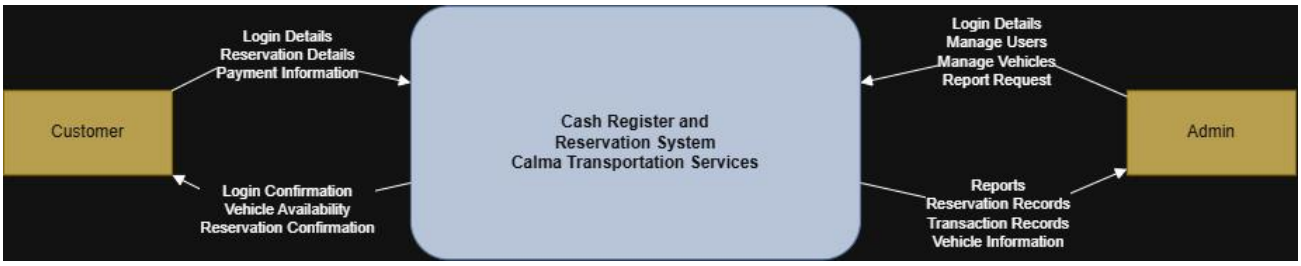
The results of the evaluation process may also provide valuable recommendations for future enhancements. Feedback obtained from users can help identify opportunities to improve system features, increase efficiency, strengthen security measures, and enhance overall performance.

Furthermore, evaluation ensures accountability within the development process by demonstrating that the system has been properly tested and assessed before implementation. This contributes to the credibility of the study and supports the validity of the research findings.

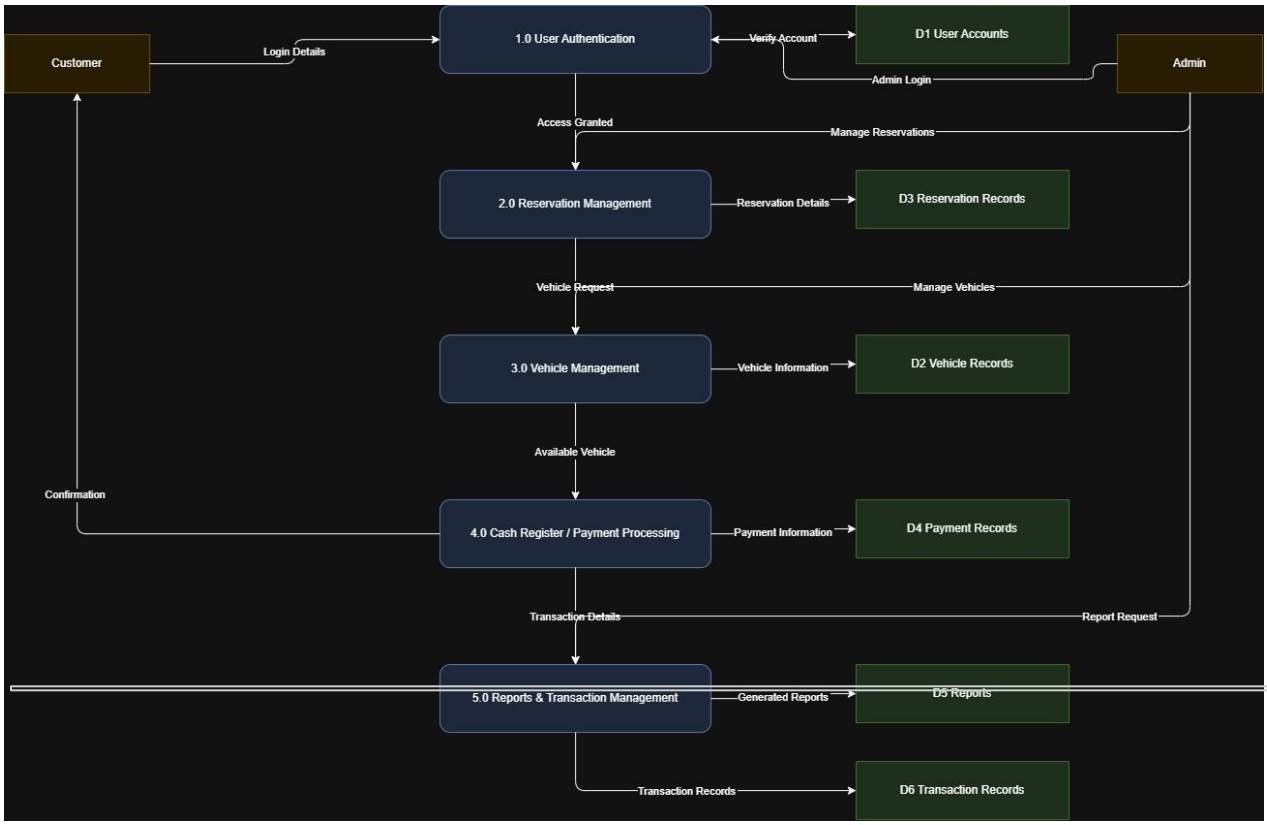
Overall, the evaluation of the proposed Cash Register and Reservation System based on functionality, usability, and reliability provides a comprehensive assessment of its quality and effectiveness. These criteria allow the researchers to measure the system's performance from both technical and user perspectives, ensuring that the developed application is capable of supporting the operational needs of Calma Transportation Services. Through this evaluation process, the researchers can determine whether the system successfully contributes to improving reservation management, transaction processing, record keeping, and overall business efficiency.

SUPPORTING DIAGRAMS

Data Flow Diagram (DFD – Level 0)



The **Data Flow Diagram (DFD) Level 0** presents a high-level overview of how information flows within the proposed Cash Register and Reservation System for Calma Transportation Services. It illustrates the interaction between the users, the system, and the database while showing how data is processed, stored, and retrieved during various operations.



The Data Flow Diagram (DFD) Level 1 provides a more detailed representation of the processes involved in the proposed Cash Register and Reservation System for Calma Transportation Services. It breaks down the main system into specific functions such as user authentication, reservation management, vehicle management, payment processing, cash register operations, and report generation. The diagram illustrates how data flows between users, the system, and the database during each process, showing how information is collected, processed, stored, updated, and retrieved to ensure efficient reservation handling, accurate payment recording, and reliable management of business operations.

Entity Relationship Diagram (ERD)

The **Entity Relationship Diagram (ERD)** illustrates the relationships between the various data entities used within the proposed Cash Register and Reservation System. It serves as the blueprint for the database structure and demonstrates how information is organized, connected, and managed throughout the application.

The database consists of three primary entities: **Users**, **Cars**, and **Reservations**.

Users Entity

The Users entity contains information about individuals who interact with the system. This includes administrators, staff members, and customers. Important attributes stored in this entity include:

- User ID
- Name
- Email Address
- Password
- Role

Cars Entity

The Cars entity contains information regarding the vehicles available for reservation. The system stores vehicle details to help customers select appropriate transportation options and enable administrators to manage available units effectively.

Attributes include:

- Car ID
- Car Name
- Brand

- Model
- Price Per Day
- Status
- Transmission Type
- Fuel Type
- Image URL

Reservations Entity

The Reservations entity stores information related to customer bookings. Each reservation record contains details necessary for monitoring and managing reservations.

Attributes include:

- Reservation ID
- Car Name
- Start Date
- End Date
- Total Price
- Status
- Created Date
- User ID
- Payment Method
- Reference Number

This entity serves as the link between customers and vehicles, allowing the system to track reservation history and transaction records.

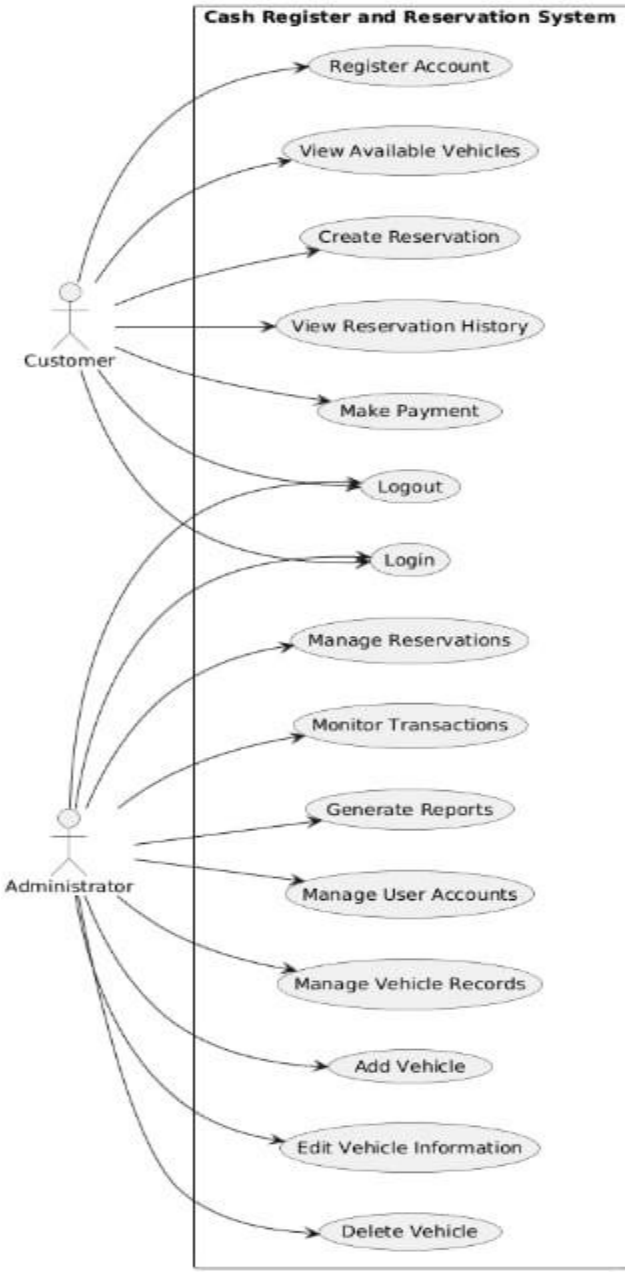
Relationships

The relationships between entities include:

- A User can create multiple Reservations.
- A Car can be associated with multiple Reservations over time.
- Administrators can manage Users, Cars, and Reservations.
- Each Reservation is associated with one User and one selected Car.

These relationships help maintain data consistency and ensure that reservation records are properly connected to both customers and vehicles.

Use Case Diagram



The **Use Case Diagram** illustrates the interactions between system users and the major functions available within the proposed Cash Register and Reservation System. It provides a visual representation of how users interact with the application and identifies the services offered by the system.

The diagram contains two primary actors:

User

The User represents customers who access the system to make reservations and manage their booking information.

The user can perform the following activities:

- Register an Account
- Login and Logout
- View Available Cars
- Create Reservations
- View Reservation History
- Make Payments through the Cash Register System

These functions allow customers to conveniently access transportation services and manage their reservations through a centralized platform.

Administrator

The Administrator is responsible for managing system operations and maintaining business records.

The administrator can perform the following activities:

- Login and Logout
- Manage Vehicle Records
- Add New Vehicles
- Edit Vehicle Information
- Delete Vehicle Records
- Manage Reservations
- Manage User Accounts
- Monitor Transactions
- Generate Reports

These administrative functions provide complete control over system operations and enable efficient management of transportation services.

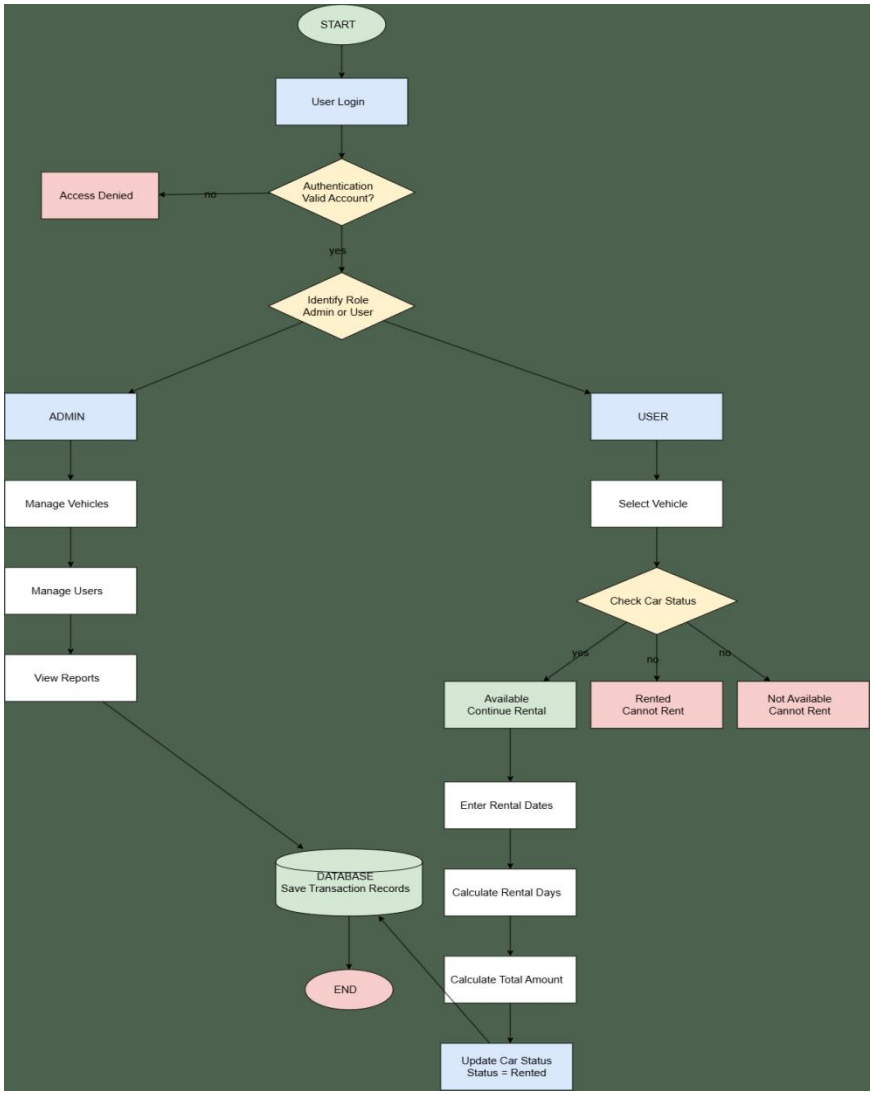
PHINMA UNIVERSITY OF ILOILO

College of Information Technology Education

CHAPTER 4

SYSTEM DEVELOPMENT, IMPLEMENTATION, TESTING, EVALUATION, RESULTS, AND DISCUSSION

4.1 System Workflow Diagram



4.2 Overview of the Developed System

The **Calma Transportation Services Cash Register and Reservation System** was developed to improve the daily operations of Calma Transportation Services by replacing its existing manual reservation, payment, and record-keeping processes with an efficient computerized system. The developed system provides a centralized platform where administrators can manage customer reservations, vehicle information, payments, and reports accurately and in real time.

Prior to the development of the system, the business relied heavily on handwritten records and manual computation of rental charges. This traditional process often resulted in booking conflicts, inaccurate billing, misplaced records, and delays in serving customers. These issues affected the overall efficiency of the business and sometimes led to customer dissatisfaction. To address these challenges, the researchers developed a web-based application that automates the major business processes involved in transportation service management.

The system allows authorized administrators to securely access the application through a login page using registered credentials. Once authenticated, administrators can perform different tasks depending on their responsibilities. The dashboard provides an overview of important information such as the number of reservations, available vehicles, ongoing rentals, completed transactions, and system notifications. This enables administrators to quickly monitor the current status of business operations without manually checking multiple records.

One of the primary features of the developed system is the **reservation management module**. This module enables administrators to register customer information, create new reservations, update booking details, and monitor the availability of vehicles. The system automatically checks whether a selected vehicle is already reserved during a specific period, preventing double bookings and scheduling conflicts. This feature improves the accuracy of reservation management while minimizing human errors commonly experienced in manual booking procedures.

The system also includes a **vehicle management module**, which stores complete information about every vehicle available for rental. Each vehicle record contains details such as the vehicle model, plate number, rental rate, seating capacity, current status, and maintenance schedule. Administrators can easily add new vehicles, update existing information, or temporarily deactivate vehicles undergoing maintenance. By maintaining updated vehicle records, the business can efficiently monitor fleet availability and reduce operational delays.

Another important component of the system is the **cash register and billing module**. Instead of manually calculating rental fees, insurance costs, penalties, and other charges, the system automatically computes the total amount based on the customer's reservation details. This minimizes mathematical errors and ensures that customers receive accurate billing information. After payment has been processed, the system generates a digital receipt that can be printed or stored electronically for future reference. Every transaction is automatically recorded in the database, making financial tracking faster and more organized.

The developed system also provides **payment management** capabilities. Administrators can record different payment methods, monitor payment status, and review previous transactions whenever necessary. Since all payment records are stored digitally, retrieving transaction history becomes significantly easier compared to searching through paper documents. This feature improves transparency and accountability in financial operations while supporting accurate bookkeeping.

4.3 System Development Methodology

The development of the **Calma Transportation Services: Cash Register and Reservation System** followed the **Agile Software Development Methodology**. Agile was selected because it provides a flexible and iterative approach to software development, allowing continuous improvement throughout the project. Instead of developing the entire system in a single phase, Agile divides the development process into smaller, manageable iterations. Each iteration focuses on developing specific features, gathering feedback, identifying issues, and implementing improvements before proceeding to the next stage. This methodology enabled the researchers to create a system that effectively addressed the operational needs of Calma Transportation Services while maintaining high software quality.

Throughout the development process, the researchers worked closely with the business owner to gather information about the existing reservation and payment procedures. Regular consultations were conducted to verify the system requirements and ensure that every feature accurately reflected the actual workflow of the business. Feedback collected during every development cycle was carefully evaluated and incorporated into succeeding revisions. This continuous collaboration helped minimize misunderstandings and ensured that the final system met both the technical requirements and business objectives.

The Agile methodology adopted in this study consisted of five major phases: **Requirements Analysis, System Design, System Development, System Testing, and Deployment**. Each phase contributed significantly to the successful completion of the system.

Requirements Analysis

The first phase involved gathering and analyzing the requirements needed for the development of the proposed system. The researchers conducted interviews and consultations with the owner and staff of Calma Transportation Services to understand the existing business processes, identify operational problems, and determine the features needed in the new system.

During this phase, the researchers examined how reservations were recorded, how payments were processed, how vehicle availability was monitored, and how customer information was managed. The researchers discovered several challenges associated with the manual system, including booking conflicts, inaccurate billing computations, difficulty retrieving customer records, delayed report preparation, and poor monitoring of vehicle maintenance schedules.

Based on the collected information, the researchers identified the functional and non-functional requirements of the system. Functional requirements included secure administrator login, customer registration, reservation management, vehicle management, automated billing, payment recording, receipt generation, maintenance scheduling, and report generation. Non-functional requirements included system security, usability,

reliability, maintainability, and acceptable system performance. These requirements served as the foundation for the succeeding development phases.

System Design

After completing the requirements analysis, the researchers proceeded with the system design phase. This phase focused on planning the overall architecture, database structure, user interface, and workflow of the application before actual coding began.

The researchers designed the system using a client-server architecture in which users interact with the application through a web browser while the server processes requests and stores information in the database. Database tables were carefully structured to organize customer information, reservation records, vehicle details, payment transactions, maintenance schedules, and administrative accounts.

The user interface was designed with simplicity and ease of use in mind. Each page was arranged to allow administrators to perform daily tasks efficiently with minimal navigation. Input validation, organized menus, clear navigation, and responsive layouts were incorporated to improve the overall user experience.

Flowcharts, system flow diagrams, database relationships, and interface prototypes were also prepared during this stage to serve as guides during software development.

System Development

The development phase involved transforming the approved system design into a fully functional web-based application. The researchers developed the frontend using **React with TypeScript** to create an interactive and responsive user interface. The backend was developed using **Node.js**, which handled business logic, user authentication, reservation processing, payment computation, and communication with the database. **MySQL** served as the database management system for storing and managing all application data.

Each module of the system was developed incrementally following Agile practices. Major modules included user authentication, dashboard management, reservation management, vehicle management, automated billing, payment processing, maintenance monitoring, and report generation.

During development, the researchers continuously integrated new features while conducting regular debugging and code refinement. Whenever issues were identified, immediate corrections were made before additional modules were implemented. This iterative process helped improve software quality while reducing the possibility of major errors during later stages of development.

System Testing

Once the core features had been developed, the system underwent extensive testing to verify that every function operated correctly and satisfied the identified requirements. Different testing methods were performed to ensure system reliability, accuracy, security, and usability.

Functional testing was conducted to verify that each module performed its intended tasks correctly. Reservation creation, billing computation, payment recording, report generation, maintenance scheduling, and user authentication were individually tested before evaluating the system as a whole.

Integration testing ensured that all modules communicated properly with one another. For example, reservation records automatically updated vehicle availability, payment transactions generated receipts, and completed rentals were reflected in system reports.

User acceptance testing (UAT) was also conducted with the business owner and selected users. Participants evaluated the usability, functionality, accuracy, and overall performance of the developed system. Feedback obtained during this phase was documented and used to make final improvements before deployment.

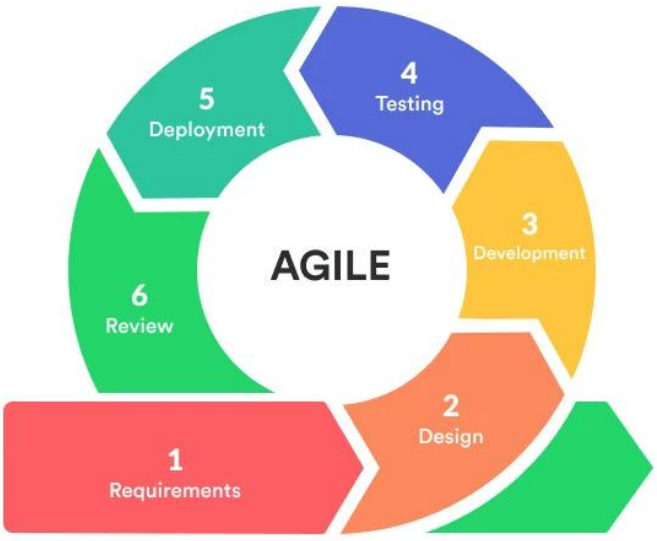
Deployment

The final phase involved deploying the completed system for actual use by Calma Transportation Services. Before deployment, the researchers performed final inspections to ensure that all modules functioned properly and that the database contained accurate information.

Administrative accounts were configured, sample operational data were entered, and users were oriented on the basic functions of the system. The researchers also demonstrated how to manage reservations, process customer payments, update vehicle records, generate reports, and monitor vehicles.

After deployment, the researchers continued to monitor system performance and addressed minor issues reported during the initial period of use. Additional adjustments were implemented whenever necessary to improve stability and usability. This continuous improvement process reflects the Agile philosophy of refining software based on user feedback even after implementation.

Overall, the Agile Software Development Methodology enabled the researchers to develop a reliable, efficient, and user-friendly Cash Register and Reservation System for Calma Transportation Services. The iterative nature of the methodology allowed continuous testing, evaluation, and enhancement throughout the project, resulting in a system that effectively addressed the operational challenges identified during the initial assessment.



4.4 System Architecture

The system architecture of the **Calma Transportation Services: Cash Register and Reservation System** follows a **client-server architecture**, which provides a centralized, secure, and efficient way of processing and managing data. This architecture enables the administrator to access the system through a web browser while the server handles application logic, processes requests, and communicates with the database. By separating the presentation layer, application layer, and data layer, the system becomes easier to maintain, update, and expand as the business grows.

The developed system consists of three primary components: the **client**, the **application server**, and the **database server**. These components work together to ensure that reservation records, customer information, payment transactions, vehicle data, and maintenance schedules are processed accurately and stored securely.



The **client side** serves as the user interface of the application. It was developed using **React with TypeScript** and allows authorized administrators to perform various operational tasks through a web browser. After logging into the system, the administrator can manage reservations, register customers, update vehicle information, process payments, monitor maintenance schedules, and generate operational reports. The interface was designed to be responsive, user-friendly, and easy to navigate, allowing users to complete tasks efficiently while minimizing errors.

The **application server** functions as the core processing unit of the system. Developed using **Node.js**, it receives requests from the client, validates user input, executes business rules, performs calculations, and manages communication between the user interface and the database. For example, when a reservation is submitted, the server verifies vehicle availability, calculates rental charges, records the transaction, and updates the vehicle's reservation status. The server also manages user authentication, ensuring that only authorized administrators can access the system and perform sensitive operations.

The **database server** uses **MySQL** to store and organize all system data. It maintains separate tables for administrator accounts, customer records, vehicle information, reservations, payment transactions, maintenance schedules, and generated reports. Through structured database relationships, the system can quickly retrieve and update information while maintaining data consistency and integrity. Proper database design also minimizes redundant information and improves overall system performance.

The interaction among these components begins when an administrator accesses the system through a web browser and submits a request, such as creating a reservation or processing a payment. The request is transmitted to the application server, where it is validated and processed according to the system's business rules. If database access is required, the application server communicates with the MySQL database to retrieve or update the necessary records. Once processing is complete, the server sends the results back to the client interface, allowing the administrator to immediately view the updated information. This communication process occurs in real time, enabling faster transaction processing and more accurate business operations.

To protect business data, the architecture incorporates several security measures. Administrator access requires user authentication through a secure login page before any system functions can be accessed. Input validation is performed on both the client and server sides to reduce invalid entries and help protect against malicious input. Sensitive information, such as user credentials, is securely stored in the database using password hashing techniques. Access to system functions is restricted to authorized users, ensuring that confidential business information remains protected.

The architecture also supports the integration of the system's major functional modules. The **Reservation Management Module** records customer bookings and automatically updates vehicle availability. The **Vehicle Management Module** maintains detailed information about each vehicle, including its rental status and maintenance history. The **Cash Register and Billing Module** automatically computes rental fees and generates digital receipts after successful payment. The **Maintenance Monitoring Module** tracks scheduled servicing and temporarily marks vehicles as unavailable whenever maintenance is required. Finally, the **Report Generation Module** compiles reservation, payment, maintenance, and vehicle information into organized reports that assist management in monitoring daily business performance.

One of the strengths of the client-server architecture is its scalability. As Calma Transportation Services expands its operations, additional features such as online customer reservations, mobile application support, SMS or email notifications, and digital payment gateway integration can be incorporated without requiring major modifications to the existing system architecture. This flexibility ensures that the developed system can continue supporting future business requirements while maintaining reliable performance.

Overall, the client-server architecture provides a stable and efficient foundation for the Cash Register and Reservation System. It enables secure communication between users and the database, automates essential business processes, reduces manual work, improves data accuracy, and supports faster decision-making through real-time information processing. By implementing this architecture, the developed system successfully meets the operational requirements of Calma Transportation Services while providing a scalable platform for future enhancements.

4.5 Hardware and Software Requirements

The successful implementation of the **Calma Transportation Services: Cash Register and Reservation System** requires appropriate hardware and software resources. These specifications ensure that the system

performs efficiently during reservation processing, payment transactions, report generation, and database management while providing a reliable experience for both administrators and customers.

4.5.1 Hardware Requirements

Table 4.1 Hardware Requirements

Component	Minimum Specification	Recommended Specification
Processor	Intel Core i3 (10th Gen) or AMD Ryzen 3	Intel Core i5 (12th Gen) or AMD Ryzen 5 and above
Memory (RAM)	8 GB	16 GB or higher
Storage	256 GB SSD	512 GB SSD or higher
Monitor	1366 × 768 Resolution	Full HD (1920 × 1080)
Internet Connection	20 Mbps	50 Mbps or higher
Printer	Thermal or Inkjet Printer	Thermal Receipt Printer
Mobile Device (Optional)	Android 10 / iOS 15	Android 13 or iOS 17 and above

The recommended hardware configuration provides sufficient processing power to handle reservation management, payment processing, receipt printing, vehicle monitoring, and report generation with minimal delays while ensuring smooth system performance.

4.5.2 Software Requirements

Table 4.2 Software Requirements

Software	Purpose
Windows 11	Operating System
Visual Studio Code	Source Code Editor
Node.js	Backend Development Environment
React with TypeScript (Vite)	Frontend Development
Bootstrap	User Interface Design
MySQL	Database Management System
XAMPP	Local Web Server and MySQL Management
Google Chrome	Web Browser for System Access and Testing
Git	Version Control
GitHub	Source Code Repository and Collaboration

The selected software tools were chosen based on their reliability, compatibility, ease of development, and strong community support. Together, these technologies provide a stable environment for developing, testing, deploying, and maintaining the **Calma Transportation Services: Cash Register and Reservation System**,

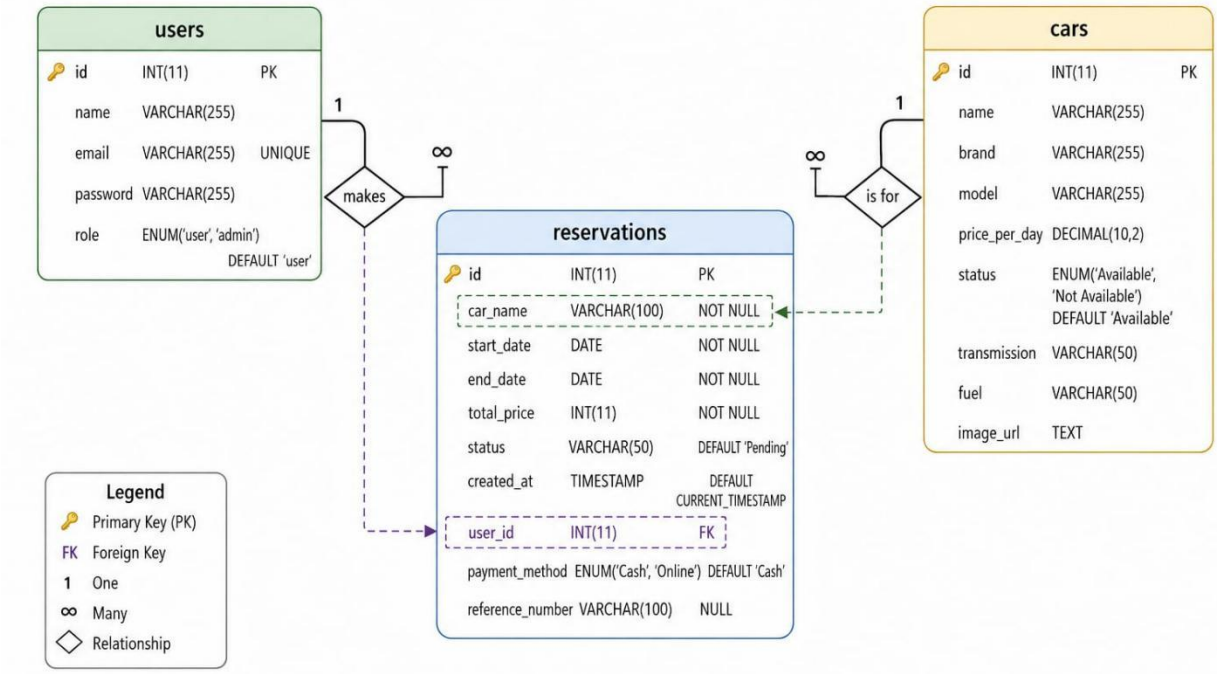
ensuring efficient reservation management, secure payment processing, accurate record keeping, and dependable system performance.

4.6 Database Design

The database serves as the foundation of the **Calma Transportation Services: Cash Register and Reservation System** by organizing and storing customer information, vehicle records, reservation details, payment transactions, cash register records, maintenance schedules, and administrative data. A well-designed database ensures efficient data storage and retrieval, minimizes data redundancy, and maintains the accuracy and integrity of information throughout the system's operations.

The researchers employed a **relational database model** using **MySQL** as the Database Management System (DBMS). Database normalization techniques were applied to eliminate unnecessary data duplication, maintain consistency among related tables, and establish proper relationships between entities such as customers, vehicles, reservations, payments, users, and maintenance records. Primary and foreign keys were implemented to enforce data integrity and ensure accurate linking of records across the database.

This database design enables the system to efficiently process reservations, manage vehicle availability, record payment transactions, generate sales and reservation reports, and support the daily operations of Calma Transportation Services while ensuring data security, reliability, and scalability for future enhancements.


users
id INT(11) PK
name VARCHAR(255)
email VARCHAR(255) UNIQUE
password VARCHAR(255)
role ENUM('user', 'admin') DEFAULT 'user'
1
makes
∞
cars
id INT(11) PK
name VARCHAR(255)
brand VARCHAR(255)
model VARCHAR(255)
price_per_day DECIMAL(10,2)
status ENUM('Available', 'Not Available') DEFAULT 'Available'
transmission VARCHAR(50)
fuel VARCHAR(50)
image_url TEXT
1
is for
∞
reservations
id INT(11) PK
car_name VARCHAR(100) NOT NULL
start_date DATE NOT NULL
end_date DATE NOT NULL
total_price INT(11) NOT NULL
status VARCHAR(50) DEFAULT 'Pending'
created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
user_id INT(11) FK
payment_method ENUM('Cash', 'Online') DEFAULT 'Cash'
reference_number VARCHAR(100) NULL
Legend
Primary Key (PK)
FK Foreign Key
1 One
∞ Many
Relationship

CHAPTER 5

CONCLUSION AND RECOMMENDATIONS

5.1 Introduction

This chapter presented the conclusions and recommendations derived from the development and implementation of the Calma Transportation Services: Implementation of a Cash Register and Reservation System. It summarized the significant findings of the study, determined whether the research objectives were achieved, and discussed the overall contribution of the developed system to the operations of Calma Transportation Services. Furthermore, this chapter provided recommendations for future system enhancements, implementation strategies, and possible research directions to further improve the efficiency and sustainability of the developed system.

5.2 Summary of Findings

The study successfully developed and implemented a **Cash Register and Reservation System** for Calma Transportation Services to address the challenges associated with manual reservation processing, payment recording, customer information management, and report generation. The implementation of the system automated the company's daily operations, resulting in improved efficiency, accuracy, and reliability.

Based on the development, implementation, and evaluation of the system, the following findings were obtained:

- The developed system successfully computerized the reservation process, enabling administrators to record, update, and monitor customer reservations efficiently.
- The system effectively monitored vehicle availability, thereby minimizing scheduling conflicts and preventing double reservations.
- The integrated cash register module accurately computed rental fees, recorded customer payments, and generated electronic transaction records, reducing errors associated with manual computations.
- The centralized database securely stored customer information, vehicle records, reservation details, and payment transactions, allowing faster retrieval and better organization of business records.
- The reporting module generated reservation summaries, payment reports, and transaction histories that supported management in monitoring business performance and making informed decisions.
- The developed system provided a user-friendly interface that simplified system navigation, reduced processing time, and improved the overall efficiency of daily operations.

Overall, the findings demonstrated that the developed system successfully achieved its intended purpose by replacing manual processes with an integrated computerized solution that enhanced operational efficiency and service quality.

5.3 Conclusion

Based on the findings of the study, it was concluded that the Calma Transportation Services: Implementation of a Cash Register and Reservation System was a practical and effective solution to the operational challenges experienced by the company. The developed system successfully automated the reservation process, customer information management, payment processing, vehicle monitoring, and report generation, thereby improving the overall efficiency of business operations.

The implementation of the system significantly reduced the limitations of the previous manual process, including inaccurate record keeping, computation errors, misplaced documents, and reservation conflicts. The use

of a centralized database ensured better organization, consistency, and security of business information while allowing authorized users to retrieve records quickly and accurately.

Furthermore, the developed system improved the quality of customer service by providing faster reservation processing, accurate payment recording, and efficient monitoring of vehicle availability. The integrated cash register module also strengthened financial accountability by maintaining complete and organized transaction records.

The objectives of the study were successfully achieved. The developed system demonstrated that integrating reservation management and cash register functionalities into a single web-based application effectively improved operational efficiency, reduced manual workload, enhanced customer service, and supported better business decision-making. The study also demonstrated the importance of adopting computerized information systems in modernizing transportation service operations.

5.4 Recommendations

Although the developed system successfully met its intended objectives, several recommendations were proposed to further improve its functionality, usability, and long-term sustainability.

System Improvements

Future versions of the system were recommended to include additional features that would further enhance business operations. These improvements included the integration of online payment gateways such as GCash, Maya, and credit or debit card payment options to provide customers with more convenient payment methods.

The researchers also recommended the implementation of SMS and email notification services to automatically send reservation confirmations, payment receipts, booking reminders, and vehicle return notifications.

A mobile application was likewise recommended to allow administrators and customers to access the system more conveniently through smartphones and tablets.

The integration of GPS vehicle tracking was also recommended to improve fleet monitoring and enhance vehicle management.

To strengthen data security, future versions of the system were recommended to implement multi-factor authentication, enhanced password encryption, comprehensive audit logs, and regular security updates.

Additional report export options, including PDF, Microsoft Excel, and CSV formats, were likewise recommended to improve documentation and report sharing.

Finally, the implementation of automated database backup and disaster recovery mechanisms was recommended to ensure data security and business
continuity.

Future Research Directions

Future researchers were encouraged to expand the scope of the study by incorporating more advanced technologies into the system.

The integration of artificial intelligence and predictive analytics was recommended to forecast customer demand and vehicle utilization more accurately.

Researchers were also encouraged to develop business intelligence dashboards capable of providing deeper insights into customer behavior, reservation trends, and financial performance.

Future studies could also investigate the integration of cloud computing technologies to improve system scalability, accessibility, and remote data management.

Moreover, predictive vehicle maintenance modules could be incorporated to monitor maintenance schedules and minimize unexpected vehicle breakdowns.

Comparative studies involving similar transportation companies were likewise recommended to further evaluate the effectiveness and adaptability of computerized reservation and cash register systems in different operational environments.

Implementation Strategies

To maximize the effectiveness of the developed system, several implementation strategies were recommended.

Comprehensive user training should be conducted before full system deployment to ensure that administrators and employees become familiar with the system's features and functionalities.

Standard operating procedures should also be established to guide users in reservation processing, payment recording, report generation, and database management.

Regular software maintenance, database backups, and system updates should be performed to maintain system reliability, security, and optimal performance.

A qualified system administrator should likewise be assigned to monitor system performance, manage user accounts, perform routine maintenance, and address technical concerns.

Finally, continuous collection of user feedback was recommended to identify opportunities for further system enhancement and ensure that future improvements aligned with the operational needs of Calma Transportation Services.

5.5 Impact of the Study

The implementation of the **Calma Transportation Services: Implementation of a Cash Register and Reservation System** produced significant benefits for the company, its employees, customers, and future researchers.

For Calma Transportation Services, the developed system improved operational efficiency by automating reservation management, payment processing, and business record management. The system reduced manual workloads, minimized processing errors, and enabled faster retrieval of business information.

Employees benefited from simplified workflows, reduced paperwork, and improved productivity, allowing them to perform daily tasks more efficiently.

Customers also benefited from faster reservation processing, accurate booking information, organized payment transactions, and improved service quality, resulting in a more convenient and reliable customer experience.

From an academic perspective, the study demonstrated the practical application of information technology in solving real-world business problems within the transportation industry. The project served as a

valuable reference for future researchers interested in reservation systems, cash register systems, transportation management, and business process automation.

Overall, the study highlighted the importance of digital transformation in improving operational efficiency, strengthening business management, and enhancing customer satisfaction through the implementation of modern information systems.

5.6 Summary

This chapter presented the conclusions and recommendations of the study. The findings demonstrated that the developed **Cash Register and Reservation System** successfully achieved its intended objectives by automating reservation management, payment processing, customer record management, vehicle monitoring, and report generation for Calma Transportation Services. The implementation of the system improved operational efficiency, minimized manual errors, strengthened data security, and supported more effective business decision-making.

Although the developed system successfully addressed the current operational requirements of the company, continuous enhancement was recommended to ensure long-term sustainability and adaptability to future technological developments. The proposed recommendations, including online payment integration, mobile application development, notification services, advanced reporting features, and intelligent analytics, were expected to further improve the system's overall functionality and effectiveness.

In conclusion, the study demonstrated that the implementation of a web-based **Cash Register and Reservation System** was an effective technological solution that modernized business operations, enhanced customer service, and supported the long-term growth and competitiveness of Calma Transportation Services.

REFERENCES

[1] M. N. Osman, N. Md. Zain, Z. Paidi, K. A. Sedek, M. N. Yusoff, and M. Maghribi, "Online Car Rental System Using Web-Based and SMS Technology," *Journal of Computing Research and Innovation*, vol. 2, no. 3, pp. 33–38, 2017. [Online]. Available: https://jcrinn.com/index.php/jcrinn/article/view/51

[2] N. A. N. Azmi and N. Hassan, "Automobile Rental System," *Applied Information Technology and Computer Science*, vol. 5, no. 1, pp. 1292–1309, 2024. [Online]. Available: https://periodical.uthm.edu.my/index.php/aitcs/article/view/12001

[3] F. V. Harissa, "Design and Prototype Implementation of a Web-Based Car Rental Information System using Agile Scrum," INOVTEK Polbeng - Seri Informatika, vol. 10, no. 3, 2025, doi: 10.35314/wfaayr38. [Online]. Available: https://doi.org/10.35314/wfaayr38

[4] Y. Yang, A. Ceder, W. Zhang, and H. Tang, "Unconstrained Estimation of Multitype Car Rental Demand," Appl. Sci., vol. 11, no. 10, p. 4506, May 2021, doi: 10.3390/app11104506. [Online]. Available: https://doi.org/10.3390/app11104506

[5] C. Caballero-Gil, P. Caballero-Gil, J. Molina-Gil, and N. García-Moreno, "Ethereum-based decentralized car rental system," Logic Journal of the IGPL, vol. 30, no. 6, pp. 926–941, Dec. 2022, doi: 10.1093/jigpal/jzac003. [Online]. Available: https://doi.org/10.1093/jigpal/jzac003

[6] J. T. Ogbiti and W. Aaron, "Development of a Web-Based Car Rental Management System," *Science World Journal*, vol. 19, no. 3, pp. 797–807, Oct. 2024, doi: 10.4314/swj.v19i3.27. [Online]. Available: https://scienceworldjournal.org/article/view/24033

[7] M. Muttaqin, F. Wadly, and P. Veranda, "Digital Transformation of Car Rental Services Using a Web Based Information System," *Journal of Information Technology, Computer Science and Electrical Engineering*, vol. 2, no. 3, pp. 127–134, 2025. [Online]. Available: https://ysmk.org/ejournal/index.php/jitcse/article/view/245

[8] W. F. W. A. Rahman and N. I. N. B. M. Rusli, “The Development of a Web-based Car Rental Business Management System,” *Journal of Mathematics and Computing Science*, vol. 9, no. 2, pp. 101–108, 2023, doi: 10.24191/jmcs.v9i2.513. [Online]. Available: https://journal.uitm.edu.my/ojs/index.php/JMCS/article/view/513

[9] A. Oktaviani and Y. S. Sari, “Web-Based Car Rental Information System (Case Study: CV. Rafael Trans),” *Jurnal Ilmiah Universitas Satya Negara Indonesia*, vol. 3, no. 2, pp. 60–66, 2025, doi: 10.59408/jisni.v2i2.3. [Online]. Available: https://ejournal.usni.ac.id/index.php/jisni/article/view/3

[10] A. H. A. N. Karsa, O. Wahyuningsih, R. Jannah, and N. Saebah, “Web-Based Car Rental Information System at CV. Mandarental Cirebon,” *Asian Journal of Engineering, Social and Health*, vol. 3, no. 1, pp. 374–380, 2024, doi: 10.46799/ajesh.v3i2.246. [Online]. Available: https://ajesh.ph/index.php/gp/article/view/246

[11] D. Sari and M. A. Jihad, “Web-Based Car Rental Service Portal Information System,” *Jurnal Mandiri*, vol. 14, no. 2, 2024, doi: 10.35335/mandiri.v14i2.467. [Online]. Available: https://ejournal.isha.or.id/index.php/Mandiri/article/view/467

[12] S. Wana, A. Candra, and J. Iryani, “Web-Based Car Rental Information System Using WhatsApp Gateway at Sasha Car Rental,” *Journal Informatika, Multimedia, and Information*, vol. 2, no. 2, pp. 41–55, 2024, doi: 10.61912/lajutek.v2i2.106. [Online]. Available: https://doi.org/10.61912/lajutek.v2i2.106

[13] A. M. S. Ahmed and M. A. M. Yunus, “AL-WEFAQ Car Rental Reservation System: Car Reservation System,” *Applied Information Technology and Computer Science*, vol. 4, no. 2, pp. 713–726, 2023. [Online]. Available: https://periodical.uthm.edu.my/index.php/aitcs/article/view/12321

[14] M. S. Jadhav and M. M. Karande, “Online Car Rental System,” *International Journal of Novel Research and Development*, vol. 8, no. 4, pp. h8–h13, Apr. 2023. [Online]. Available: https://www.ijnrd.org/papers/IJNRD2304702.pdf

[15] M. Bilal and I. F. Hanif, “Rancang Bangun Aplikasi Penyewaan Kendaraan Berbasis Web Menggunakan Metode Waterfall,” *Journal of Information System Research (JOSH)*, vol. 6, no. 4, pp. 1956–1963, 2025, doi: 10.47065/josh.v6i4.7890. [Online]. Available: https://ejurnal.seminar-id.com/index.php/josh/article/view/7890

[16] N. A. A. Bakar and M. F. Yuspani, “Initial stage in developing an online car rental system (OCRS) using customization business model,” *Journal of Physics: Conference Series*, vol. 1529, no. 2, Art. no. 022033, 2020, doi: 10.1088/1742-6596/1529/2/022033. [Online]. Available: https://doi.org/10.1088/1742-6596/1529/2/022033

[17] K. Padhiyar and M. K. Rathod, “Implementation of Car Management and Rental System: Web based Approach,” *International Journal of Innovative Research in Science, Engineering and Technology*, vol. 14, no. 3, 2025, doi: 10.15680/IJIRSET.2025.1403451. [Online]. Available: https://www.ijirset.com/upload/2025/march/451_Implementation.pdf

[18] T. Gunasekaran and S. H. Arbain, “DriveEasy: Car Rental Management System,” *Applied Information Technology and Computer Science*, vol. 6, no. 1, pp. 861–880, 2025. [Online]. Available: https://periodical.uthm.edu.my/index.php/aitcs/article/view/16292

[19] J. R. P. Soares, J. E. da Costa, and J. F. A. Bernardo, “Implementation of Car Rental Management Application Systems Based on Website Used PHP and MySQL (Case Study: Kailelo Car Rental),” *Dili Journal of Engineering, Technology and Innovation*, vol. 1, no. 1, 2025. [Online]. Available: https://dilijetin.org/index.php/dilijetin/article/view/1

[20] K. Waje, A. Malekar, A. Patil, A. Shelawale, and P. Tiware, “Car Rental System,” *International Journal of Science and Social Science Research*, vol. 2, no. 1, pp. 103–110, 2024, doi: 10.63671/ijsssr.v2i1.240. [Online]. Available: https://www.ijsssr.com/index.php/j/article/view/240

[21] P. Nahnisha, B. Jai Sai Anvitha, and R. Prema, “Online Car Rental System,” *International Journal of Computer Science and Mobile Computing*, vol. 11, no. 3, pp. 102–107, Mar. 2022, doi: 10.47760/ijcsmc.2022.v11i03.012. [Online]. Available: https://ijcsmc.com/docs/papers/March2022/V11I3202226.pdf

[22] A. D. Philogin and A. Ram Prasath, “Revolutionizing Car Rental: A Cutting-Edge Online Administration System,” *Tuijin Jishu/Journal of Propulsion Technology*, vol. 44, no. 6, 2023. [Online]. Available: https://www.propulsiontechjournal.com/index.php/journal/article/view/5605

[23] P. X. Qi, A. S. Shibghatullah, and K. Subaramaniam, “Rehabilitating Flood-Damaged Cars for Sustainable Car Rental Services: A Web-Based System,” *Proceedings of the International Conference on Artificial Life and Robotics*, vol. 29, pp. 1056–1060, 2024, doi: 10.5954/ICAROB.2024.GS7-1. [Online]. Available: https://doi.org/10.5954/ICAROB.2024.GS7-1

[24] M. G. Albino and V. Acebedo, “Development of Car Rental Management System with Scheduling Algorithm,” *International Journal of Science, Engineering and Technology*, vol. 9, no. 2, pp. 1–7, 2021. [Online]. Available: https://www.ijset.in/wp-content/uploads/IJSET_V9_issue2_162.pdf

[25] M. A. T. Golo and R. E. Encarnacion, “Revolutionizing the Rental Services in Siargao Island: Basis for Developing an Online Vehicle Rental Management System,” *International Journal of Advanced Research in Science, Communication and Technology*, vol. 4, no. 1, pp. 438–447, Jun. 2024, doi: 10.48175/IJARSCT-18754. [Online]. Available: https://www.ijarsct.co.in/Paper18754.pdf

[26] M. F. V. De Jesus, R. B. Y. Dela Cruz, B. F. D. Marcelo, D. L. Valencia, P. K. F. Yumul, and J. P. Valete, “PARAM: A Web-based Booking and Management System with Mobile Tracking Application for Param Car Rentals,” *MCC-IT Research Journal*, vol. 1, no. 1, 2021. [Online]. Available: https://mcc.edu.ph/academic-research/download/MCC-IT%20Research%20Journal%202020-2021.pdf

[27] B. G. S. Grepon, N. T. Baran, K. M. V. C. Gumonan, A. L. M. Martinez, and M. L. E. Lacsa, “Designing and Implementing e-School Systems: An Information Systems Approach to School Management of a Community College in Northern Mindanao, Philippines,” *International Journal of Computing Sciences Research*, vol. 6, pp. 792–808, 2022, doi: 10.25147/ijcsr.2017.001.1.74. [Online]. Available: https://stepacademic.net/ijcsr/article/download/255/103/

[28] M. P. Marafo, J. S. Antonio, B. A. Palay-en, C. M. Toribio, and G. S. Guaki, “Arkila: A Progressive Web App Car Rental Management System for Kataguan Rides in La Trinidad, Benguet,” *Southeast Asian Journal of Science and Technology*, vol. 10, no. 1, pp. 82–98, 2025. [Online]. Available: https://sajst.pti.edu.ph/online/index.php/sajst/article/view/359/311

[29] K. B. Estoque, M. Y. Maniri, C. J. Noval, J. D. Quijano, J. M. Rivera, A. M. Soriano, and S. Ortega, “The Role of Assurance and Safety in Shaping Customer Referrals: Insights from Car Rental Services in Malaybalay City,” *International Journal of Sustainability and Advanced Integrated Research*, vol. 2, no. 1, pp. 116–121, 2026. doi: 10.65339/ijsair.V2.I1.14. [Online]. Available: https://ijsair.com/index.php/ijsair/article/view/51/40

[30] G. M. Calosa, A. M. C. Aguila, C. A. G. Silang, and C. B. D. Sarmiento, “Transerve: A Linear Programming–Based Decision Support System for Optimizing Vehicle Reservation Operations,” in *Proceedings of the 7th Asia Pacific Conference on Industrial Engineering and Operations Management*, 2026, doi: 10.46254/AP07.20260065. [Online]. Available: https://index.ieomsociety.org/index.cfm/item/62847

[31] M. M. E. Abat, J. A. D. Buen, and V. R. N. Giron, “Online Car Rental System,” unpublished B.S. thesis, College of Information Systems, Don Mariano Marcos Memorial State University–North La Union Campus, La Union, Philippines, May 2023. [Online]. Available: https://lakasa.dmmmsu.edu.ph/handle/123456789/92

[32] E. Balintag, K. J. H. Concepcion, P. J. Dagman, J. A. Ducos, D. Fernandez, and E. Villanueva, “SORMS: Steffi’s Online Reservation Management System,” *Psychology and Education: A Multidisciplinary Journal*, vol. 36, no. 7, 2025. [Online]. Available: https://www.ejournals.ph/article.php?id=32526

[33] A. G. L. Montesines, M. C. B. Ignacio, M. H. Somera, G. D. Balbin, L. P. Tabu, R. U. Briagas, M. M. Dajan, and M. M. Caisip, “Development of an Online Reservation and Monitoring System for the Use of TWC Dormitory, Function Rooms and Vehicles,” TESDA Women’s Center, Philippines, 2017. [Online]. Available: https://twc.tesda.gov.ph/advocacy/researches/7.pdf

[34] F. V. Dalisay, “Development of a Multi-platform Online Reservation System for Car Services,” *Asian Journal of Multidisciplinary Studies*, vol. 2, no. 1, pp. 165–173, May 2019. [Online]. Available: https://ajms.pti.edu.ph/online/index.php/ajms/article/view/397

[35] E. O. Libunao, A. A. Libunao, L. P. Pornuevo, K. R. P. Sunido, J. A. G. Viloria, and V. B. Figueroa, “TrialaTS: Triala Transportation System,” *Psychology and Education: A Multidisciplinary Journal*, vol. 37, no. 5, pp. 474–481, 2025, doi: 10.70838/pemj.370507. [Online]. Available: https://doi.org/10.70838/pemj.370507

[36] M. J. Palmones and V. Maravilla Jr., “Evaluating Danao City's Transportation Sector: Towards a QR Code-Based Online Payment System,” *Psychology and Education: A Multidisciplinary Journal*, vol. 44, no. 4, pp. 446–452, 2025, doi: 10.70838/pemj.440404. [Online]. Available: https://ejournals.ph/article.php?id=31890